PCEOT502 Power Electronics

Module 1

Semi-controlled switch: SCR (Thyristor) – Symbol, Structure, Characteristics (Static and dynamic) – Turn-on and Turn-off phenomena – Ratings- Gate control of SCR – Gate pulse magnitude and duration requirements- Typical gate drive circuits – Gate synchronisation – Isolated gate drives [3 Hrs]

SILICON CONTROLLED RECTIFIER

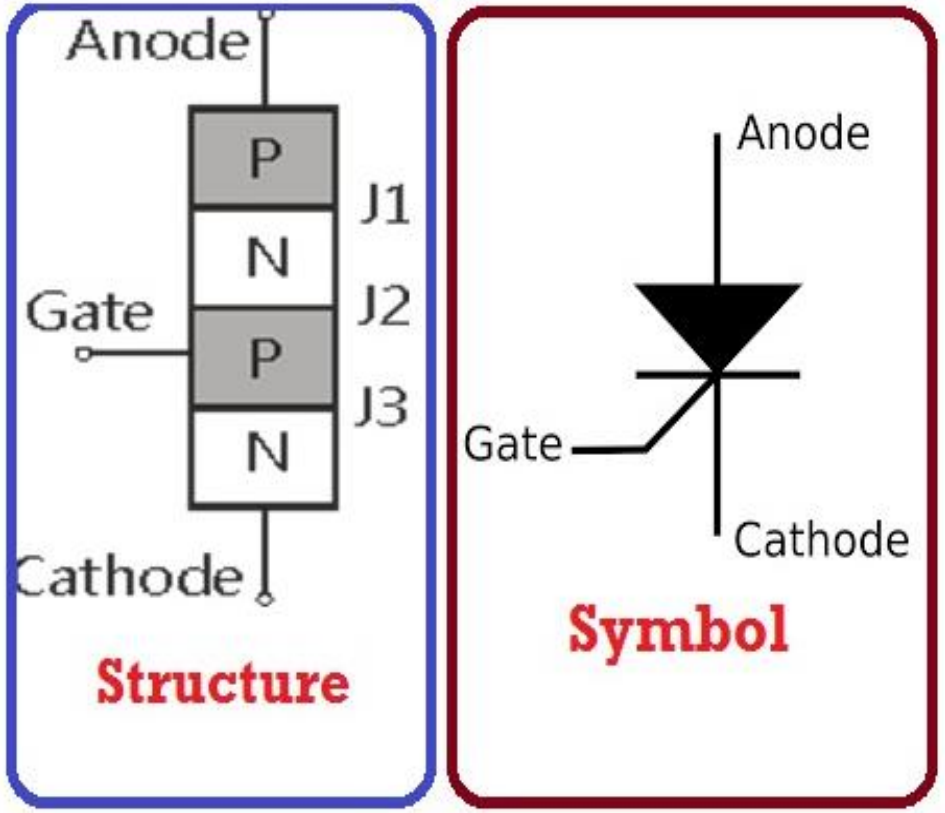
(SCR)

THYRISTORS (Contd…)

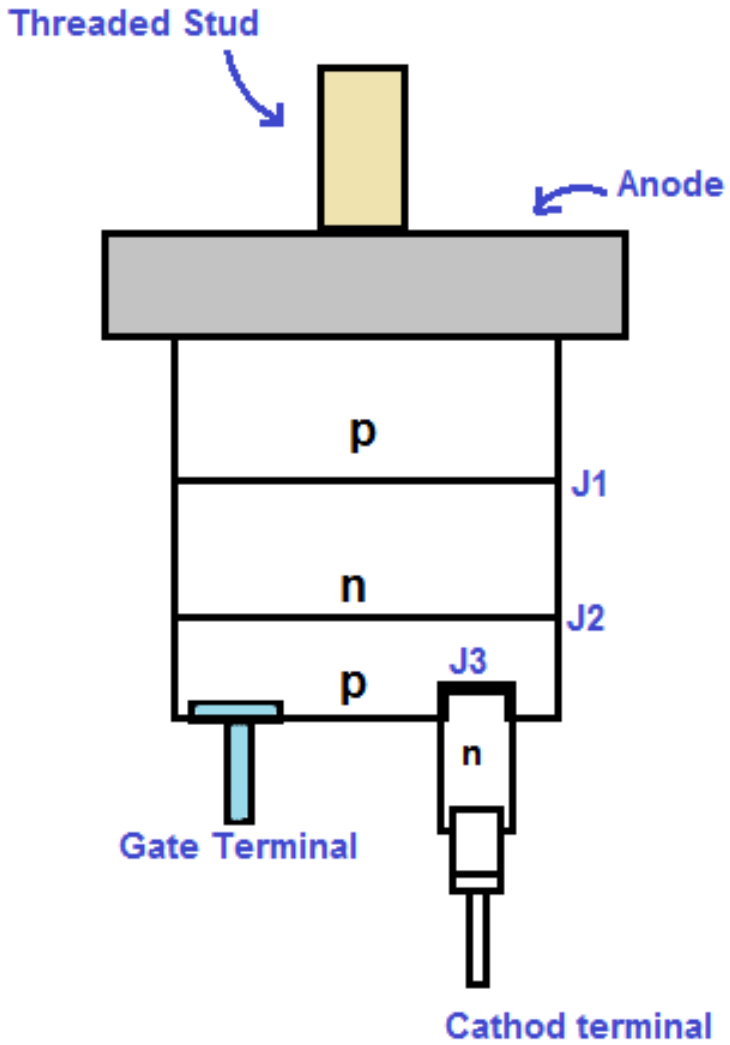
- Has a PNPN structure
- Can be viewed as a PNP and NPN transistor connected back-to-back
- THYRISTOR → THYRatron + transISTOR
- Is a device similar to a transistor and has characteristics similar to a thyratron tube
- Once the device is ON it acts like a rectifier and so it is called as SCR.
- It has common properties of diodes, resistors and transistors.
- Thyristor is a unidirectional device and conducts current in only one direction.
- It can be used only for switching and cannot be used for amplification.

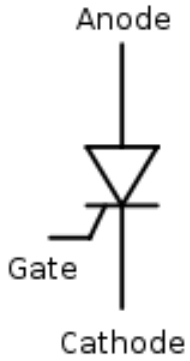
THYRISTORS -Principle

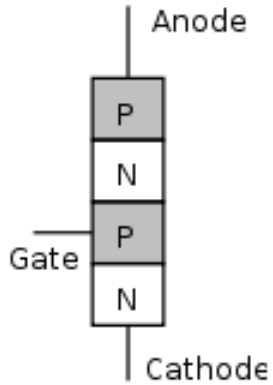
- 4 layered *PNPN* semiconductor device, having 3 junctions J_1, J_2, J_3 and 3 terminals-Anode (A), Cathode (K) and Gate (G).
- Main terminals – Anode and Cathode
- Control terminal – Gate
- Gate terminal is kept near the Cathode
- Outer p-region is the Anode
- Outer n-region is the Cathode
- Inner p-region is the Gate
- Gate terminal controls the device
- Voltage rating of 10 kV and rms current rating of 3 kA are available



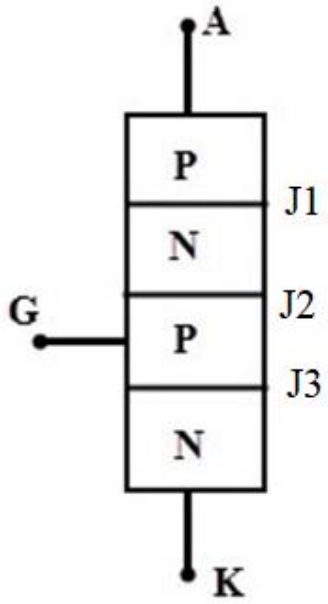
THYRISTORS -Principle



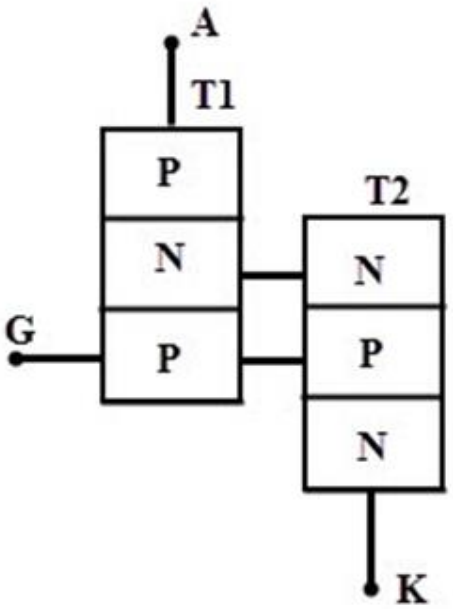




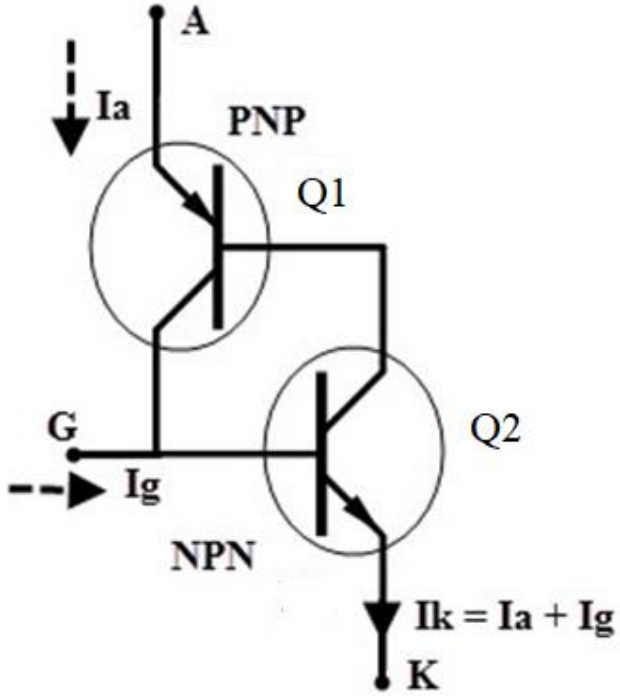
TWO TRANSISTOR MODEL



Basic Structure of SCR



Two Transistor Model



Two Transistor
Equivalent Circuit

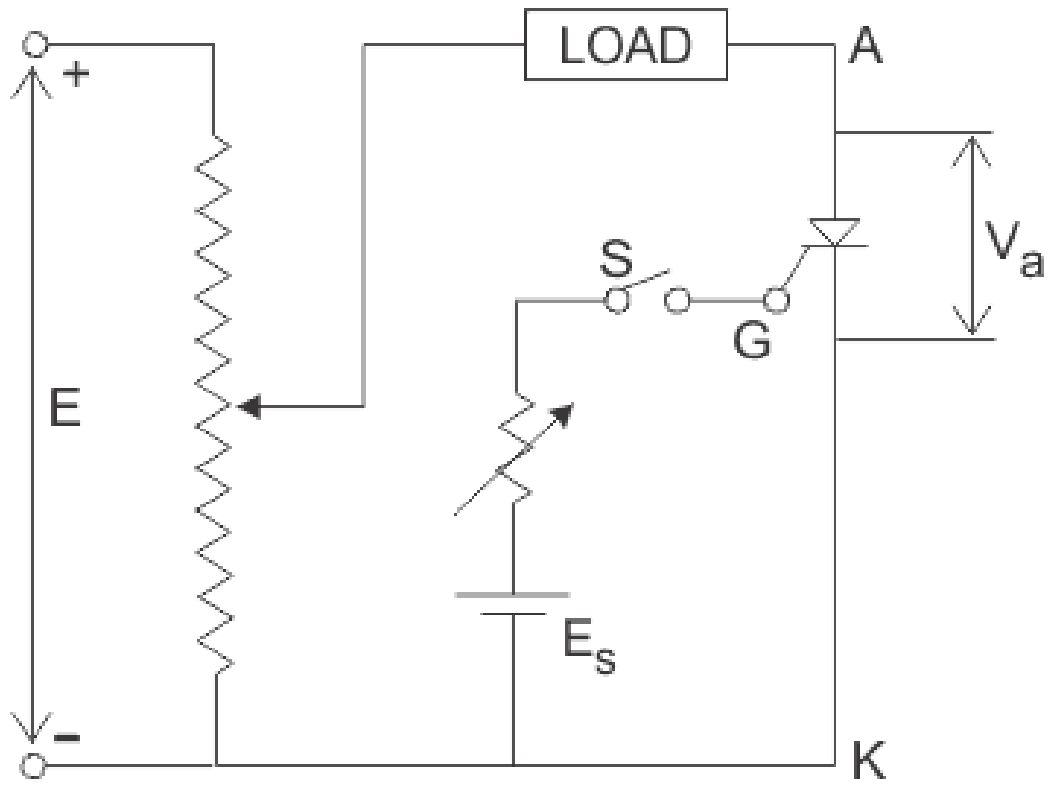
TWO TRANSISTOR MODEL

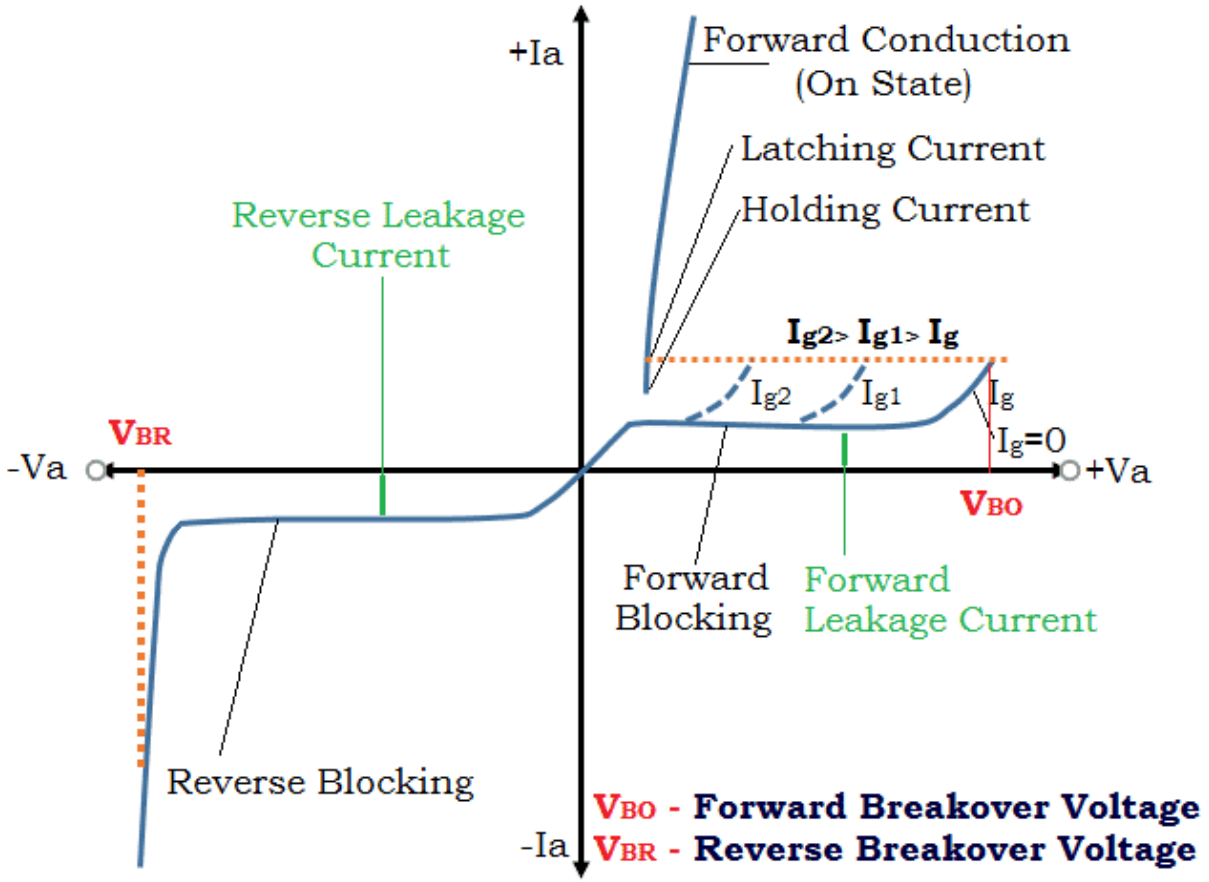
- SCR or thyristor is a three terminal semiconductor device having *P-N-P-N* structure.
- Two transistor analogy of SCR is a method of representing SCR in the form of two transistor model.
- The basic operating principle of SCR can realized by using the two transistor model.
- This represents SCR as the combination of *PNP* and *NPN* transistor.
- Anode of the SCR is connected to the emitter of the *PNP* transistor
- Cathode of the SCR is connected to the emitter of *NPN* transistor
- Both transistors are connected in common base configuration

THYRISTORS – Static V-I characteristics

- **3 modes** of operation of a thyristor –
  - reverse blocking mode
  - forward blocking mode (off-state)
  - forward conduction mode (on-state)
- Anode and cathode are connected to the main source *E* through the load
- Gate and cathode is connected to a source E_s **.**
- E_s provides a +ve gate current from gate to cathode
- Switching the SCR from forward blocking state (OFF – state) to forward conduction state (ON – state) is known as **turning-ON process of SCR. This is also called as triggering.**
- The criteria for triggering the SCR depends on the several factors like supply voltage, gate current, temperature, etc.

Static V-I characteristics (Contd….)

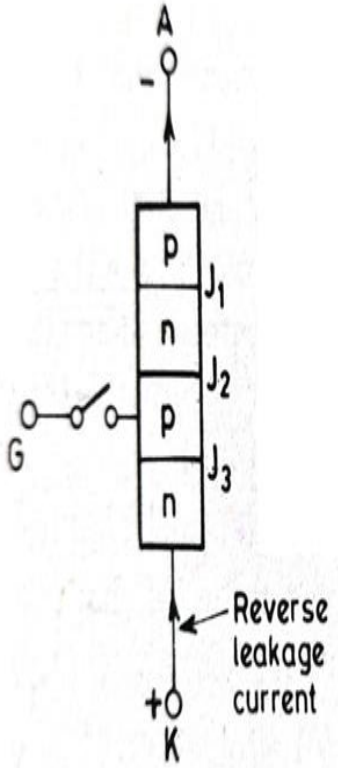


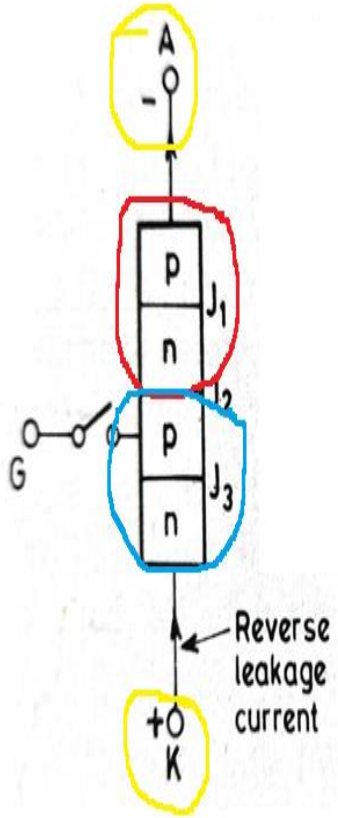


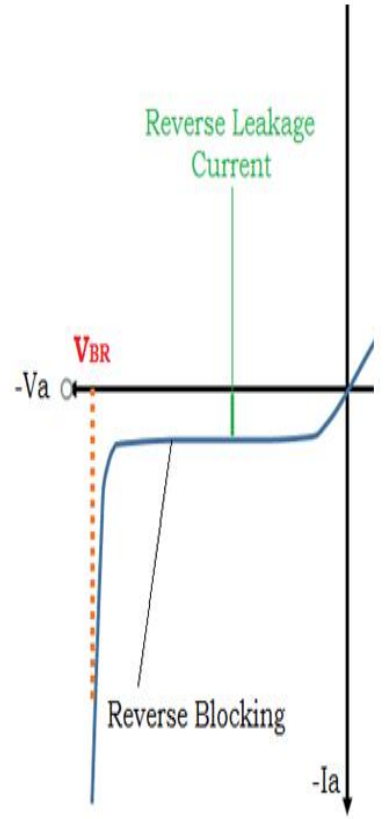
Static V-I characteristics (Contd….)

Reverse blocking mode:-

- Also called off-state of the thyristor
- When cathode is +ve wrt anode and switch S open, the thyristor is reverse biased.
- Junction J_2 is forward biased and J_1 and J_3 are reverse biased.
- Device is now similar to two diodes connected in series across a reverse voltage
- Small leakage current (of the order of a few milliamps or microamps depending on the rating of the SCR) flows.







Static V-I characteristics (Contd….)

Reverse blocking mode:- (Contd…)

- When the reverse voltage is increased an avalanche occurs at J_1 and J_3 at a critical breakdown value.
- This breakdown value is called *reverse breakdown voltage, V_{BR}.*
- Now the reverse current increases rapidly
- Large current give rise to more losses in the SCR
- May damage the thyristor as the junction temperature may exceed its permissible value.

Static V-I characteristics (Contd….)

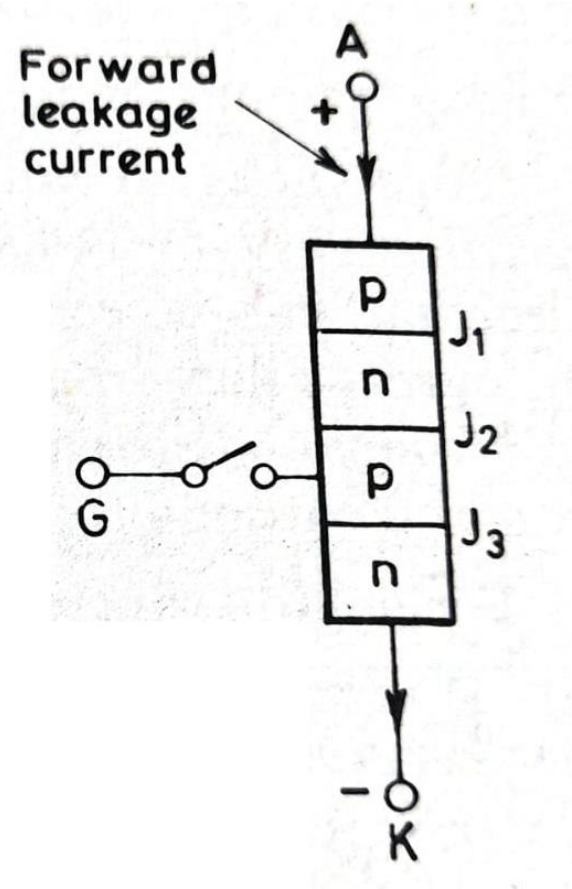
Reverse blocking mode:- (Contd…)

- So the maximum working reverse voltage should not exceed V_{BR}.
- When the applied reverse voltage is less than V_{BR} the device offers high impedance in the reverse direction.
- SCR in this mode is treated as an open switch

Static V-I characteristics (Contd….)

Forward blocking mode:-

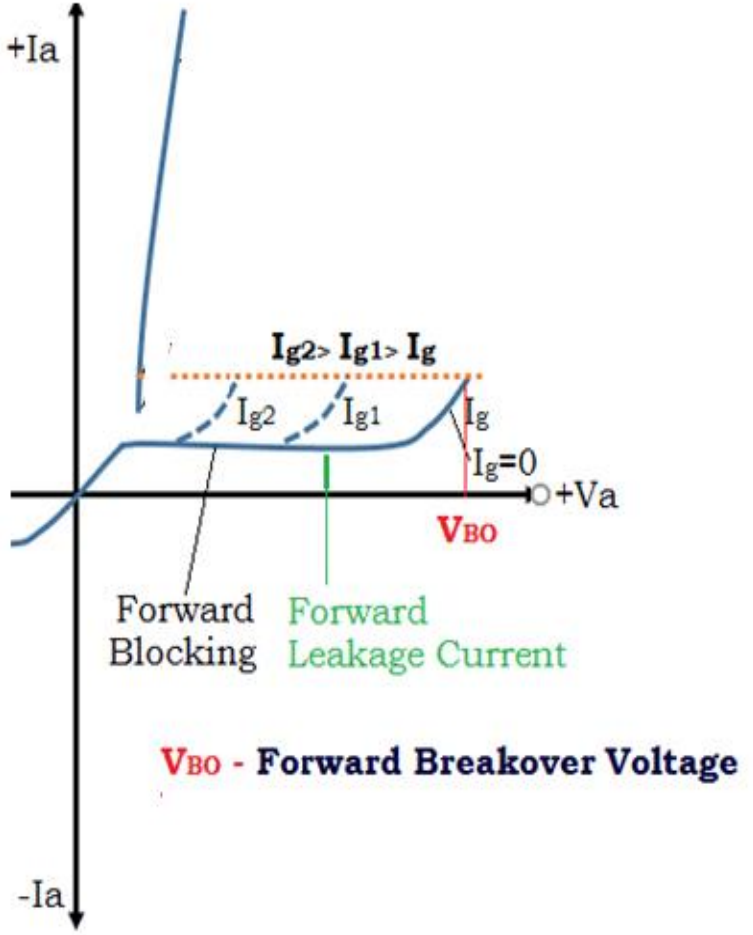
- When anode is +ve wrt cathode and switch S open, the thyristor is forward biased.
- Junction J_2 is reverse biased and J_1 and J_3 are forward biased.
- Small current flows, known as the forward leakage current.



Static V-I characteristics (Contd….)

Forward blocking mode:- (Contd...)

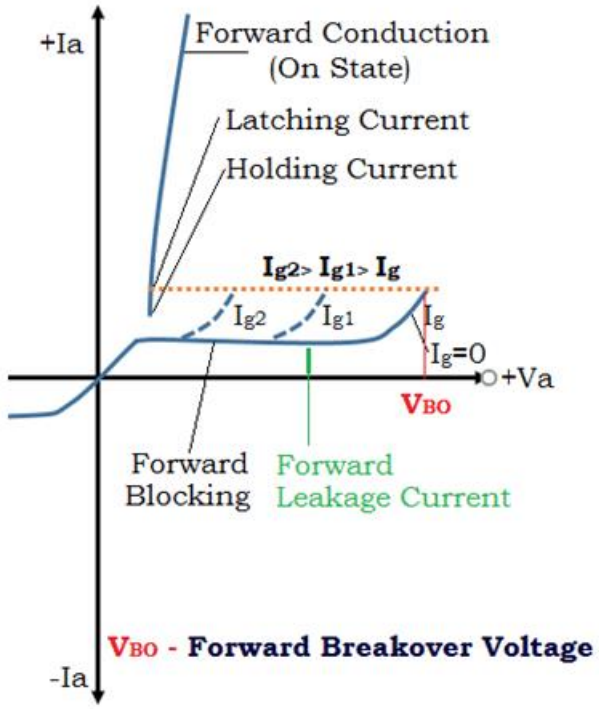
- When the forward voltage is increased the reverse biased junction J_2 will have an avalanche breakdown occurs at a critical breakdown value.
- This critical value is known as *forward breakover voltage, V_{BO}.*
- When forward voltage is less V_{BO} than SCR offer high impedance
- SCR in this mode is treated as an open switch



Static V-I characteristics (Contd….)

Forward Conduction mode:-

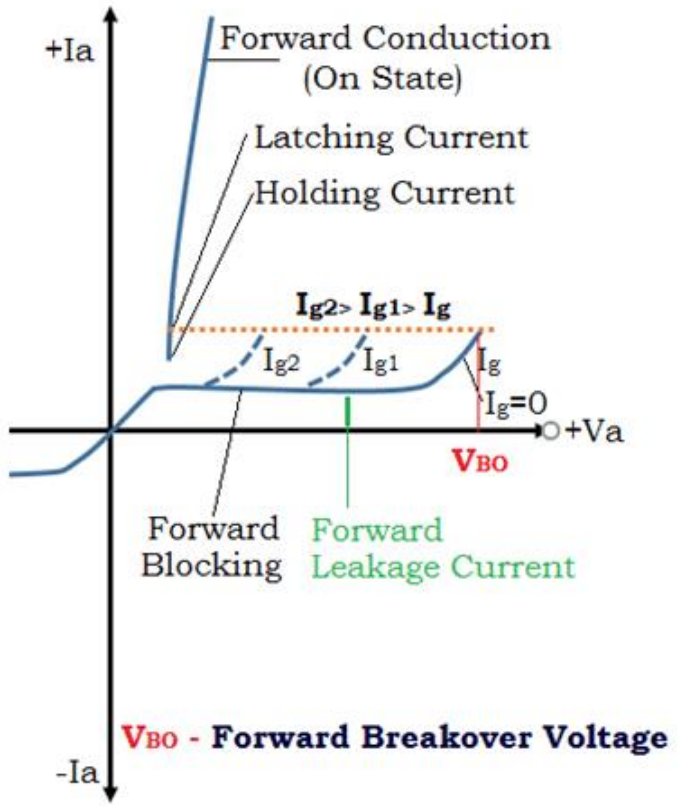
- The SCR conducts current from anode to cathode with a very small voltage drop across it.
- Thyristor is moved from forward blocking to forward conduction mode by turning it ON ***with a voltage greater than the forward breakover voltage across the SCR*** *or* ***by applying a gate pulse across gate and cathode***.
- In this mode the SCR is in the on-state
- Behaves like a closed switch



Static V-I characteristics (Contd….)

Forward Conduction mode:- (Contd....)

- On-state voltage drop is of the order of 1 to 2 volt (it depends on the rating of the SCR).
- On-state voltage drop increases slightly with the increase in anode current.
- Anode current is limited by the load impedance alone in this mode because the voltage drop across SCR is very small
- This small voltage drop is due to the ohmic drop in the four layers.
- V_{BO} *and* V_{BR} are nearly same , but V_{BR} slightly greater than V_{BO}.
- V_{BO} is taken as the final voltage rating of the device for design purpose

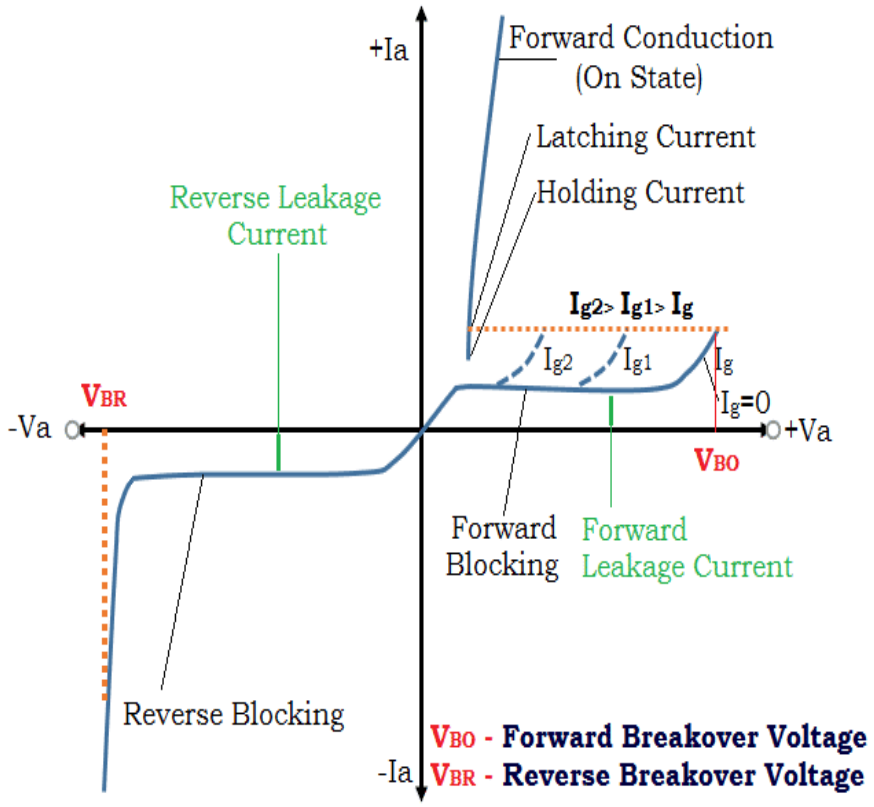


Latching current (I_l) :- *defined as the minimum value of anode current which it must attain during* ***turn-on*** *process to maintain conduction even if the gate signal is removed.*

- Once the thyristor starts conducting gate loses its control.
- The thyristor can be turned-off only if the anode current (forward current) falls below a low-level current called the holding current.

Holding current (I_h):- *defined as the minimum value of anode current below which it must fall for* ***turning-off*** *the thyristor.*

- Latching current(I_l) is higher than the holding current(I_h)
- Latching current is taken as 2 to 3 times the holding current
- For industrial applications, holding current is taken as zero, even though the typical value is 10 mA.
- Latching current – associated with turn-on process
- Holding current – associated with turn-off process



SWITCHING CHARACTERISTICS OF SCRs

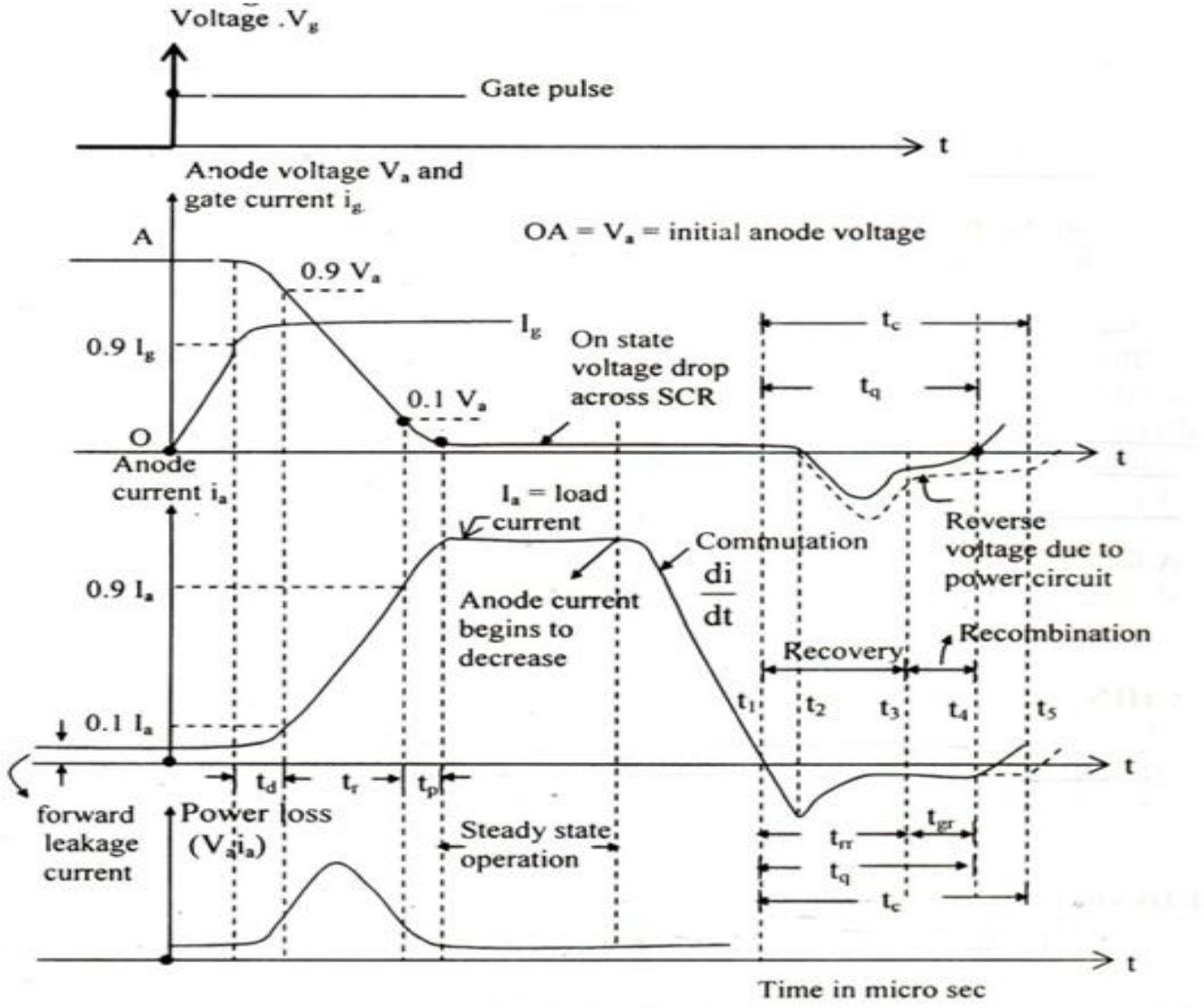
SWITCHING CHARACTERISTICS

- Static & switching chara of thyristors are required for the reliable design of converters
- Switching chara – also known as dynamic or transient chara
- During turn-on and turn-off, a thyristor is subjected to different voltages across it and different current through it.
- Switching chara – variation of voltage across a thyristor and current through it plotted against time

SWITCHING CHARACTERISTICS-during turn-on

- A forward biased thyristor is turned on by applying a +ve voltage between gate and cathode.
- There is a transition time from forward off-state to forward on-state, called **turn-on time**.
- ***Turn-on time***:- defined as the time during which it changes form forward blocking to final on-state.
- Total turn-on time (T_{on})– 3 intervals - delay time (t_d), rise time (t_r), and spread time (t_p)

SWITCHING CHARACTERISTICS
-during turn-on & turn-off



SWITCHING CHARACTERISTICS-during turn-on

Delay time (t_d) :-

- time measured from the instant at which gate current reaches $0.9I_g$ to the instant at which anode current reaches $0.1\ I_a$. ***(time from $0.9I_g$ to $0.1\ I_a$).***
- I_g & I_a are the final values of gate current & anode current.
- time during which anode voltage falls from V_a to $0.9V_a$, where V_a = initial value of anode voltage
- Time during which the anode current rises from the forward leakage current to $0.1\ I_a$.

SWITCHING CHARACTERISTICS-during turn-on

Delay time (t_d) :- (Contd...)

- In the forward blocking state, the anode voltage is OA & anode current is small leakage current.
- Turn-on process is indicated by a rise in the anode current from the small leakage current.
- At the same time there is a drop in the anode-cathode voltage from the forward blocking voltage.

SWITCHING CHARACTERISTICS-during turn-on

Delay time (t_d) :- (Contd...)

- When a gate signal is applied a gate current starts to flow from gate to cathode.
- Due to the p-layer, the gate current has a non-uniform distribution
- Its value is higher near the gate , decreases rapidly as the distance from the gate increases
- During delay time anode current flows in a narrow region near the gate.
- i.e., gate current density is highest near the gate.

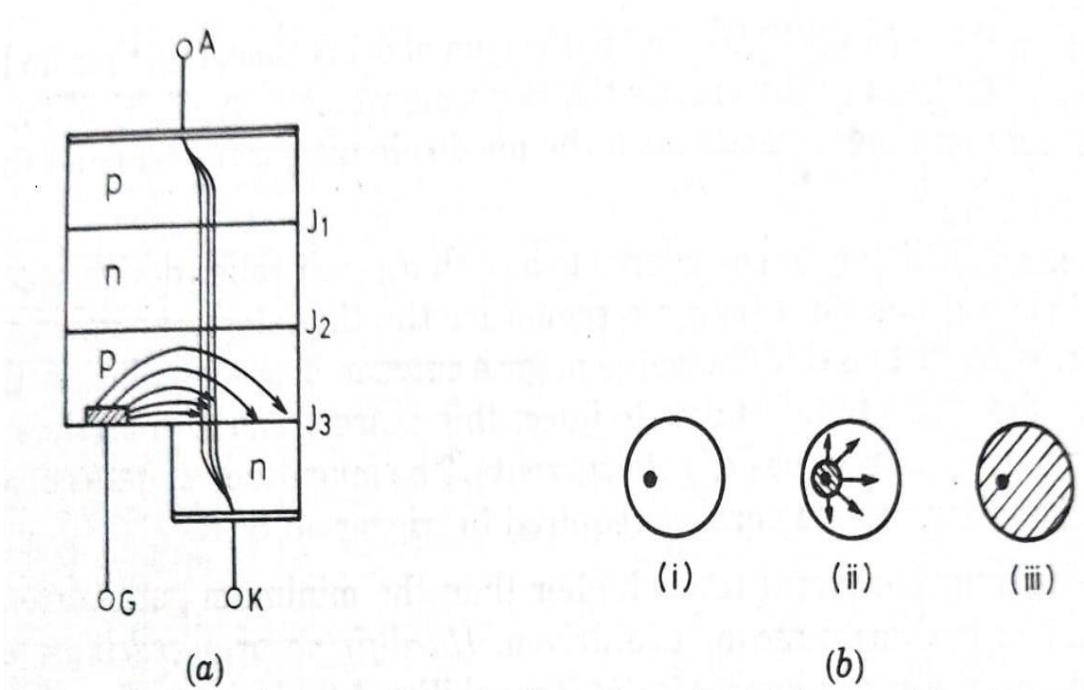


Fig. (*a*) Distribution of gate and anode currents during delay time (*b*) Conducting area of cathode (*i*) during t_d (*ii*) after t_r (*iii*) after t_p.

SWITCHING CHARACTERISTICS-during turn-on

Rise time (t_r) :-

- time taken by the anode current to rise from $0.1I_a$ to $0.9I_a$
- Time required by the forward blocking off-state voltage to fall from 0.9 to 0.1 of its initial value
- Inversely proportional to magnitude of gate current
- Can be reduced if high and steep current pulses are applied to the gate
- Nature of anode circuit mainly determines the rise time.

SWITCHING CHARACTERISTICS-during turn-on

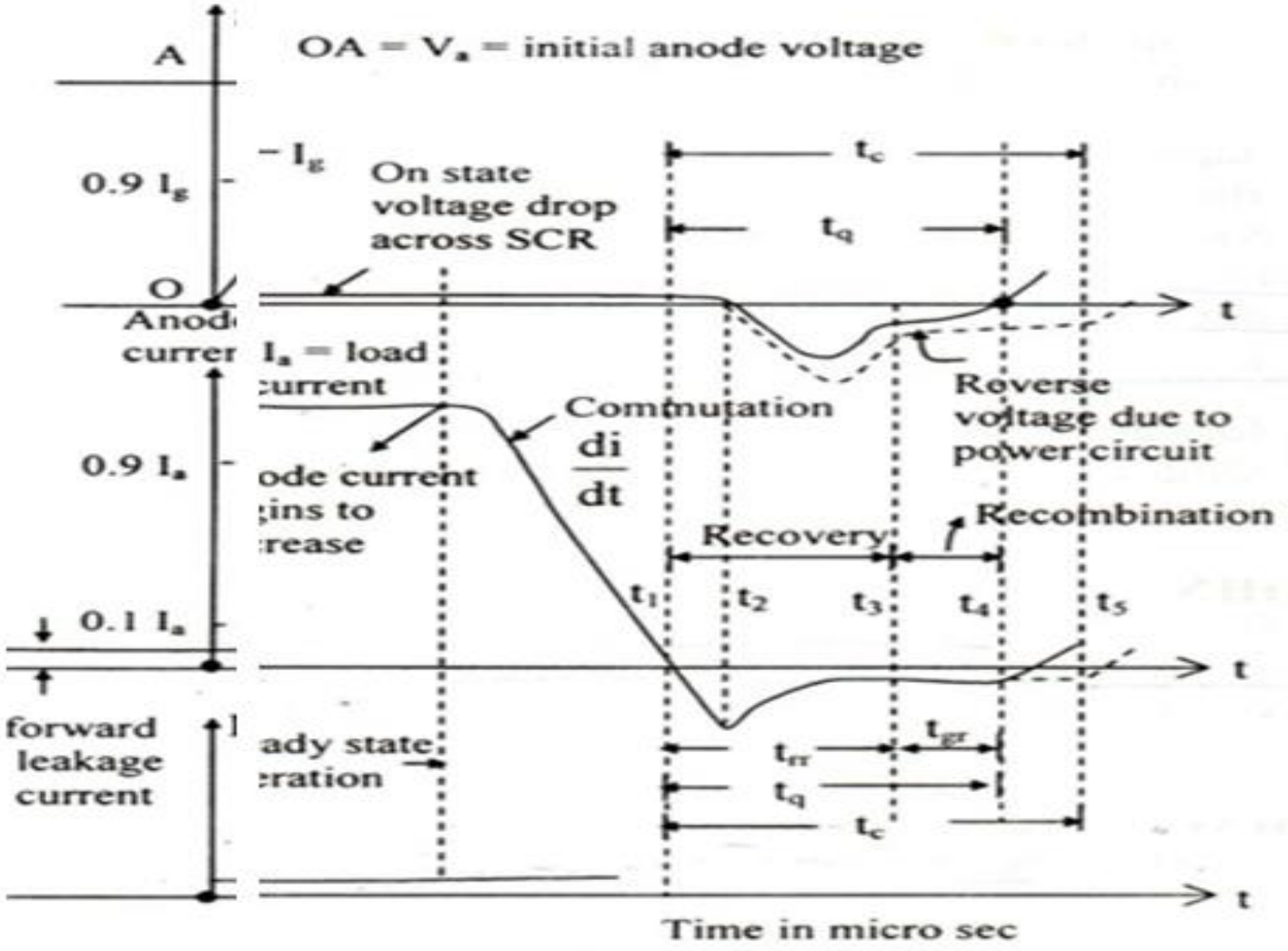
Spread time (t_p) :- (Contd...)

- time taken by the anode current to rise from $0.9I_a$ to I_a
- Time required by the forward blocking voltage to fall from 0.1 of its initial value to the on-state drop which is around 1 to 1.5V
- During t_p the conduction spreads over the entire cross-section of the cathode of SCR
- Spreading interval depends on the area of the cathode and gate structure
- After t_p anode current reaches steady state value
- Also the voltage drop across SCR is equal to the on-state voltage drop.

SWITCHING CHARACTERISTICS-during turn-on

- Total turn-on time = delay time + rise time + spread time
- Total turn-on time depends on anode circuit parameters and waveshape of the gate signal.
- Rise time is typically of the order of 1 to 4 μs
- During turn-on SCR is considered as a charge controlled device
- Certain amount of charge must be injected into the gate region for the conduction to begin.
- So higher the gate current lesser will be the time to inject this charge
- Turn-on time can be reduced by using gate currents of higher values
- Magnitude of gate current is usually 3 to 5 times the minimum gate current required to trigger an SCR.

SWITCHING CHARACTERISTICS
-during turn-off



Turn off characteristics

- The turn off time tq of a thyristor is defined as the time between the instant anode current become zero and the instant SCR regains forward blocking capability. (3 to 100µs)
- $t_q = t_{rr} + t_{gr}$.

  t_{rr} - Reverse recovery time

  t_{gr} - gate recovery time
- The reason for the reversal of anode current after t1 is due to the presence of carriers stored in the 4 layers.

- The reverse voltage surge appears across the thyristor terminals may damage it.
- This is avoided by using protective RC elements across SCR.
- At instance t3, when reverse recovery current has fallen to nearly zero value, end junctions J1 and J3 recover and SCR is able to block the reverse voltage
- J2 still has trapped charges. Therefore the thyristor is not able to block forward voltage at t3.
- Trapped charges gets decayed only by recombination.
- Recombination is possible by maintaining a reverse voltage (0 to -50V) across SCR

- The time for the recombination of charges between t3and t4 is called gate recovery time
- At t4, J2 recovers, forward voltage can be reapplied between anode and cathode
- The turn off time provided to SCR by practical circuit is called circuit turn off time, tc
- Tc must be greater than tq for reliable turn off, otherwise commutation failure (undesired turn on) may happen
- Converter grade SCR : tq - 50 to 100µs
- Inverter grade SCR : tq - 3 to 50µs

THYRISTOR- *RATINGS*

THYRISTOR RATINGS

- Different ratings are:- voltage, current, power, dv/dt, di/dt, turn-on time, turn-off time etc.
- **Representation of subscripts:-**
- First subscript indicate the direction or state: **D-** forward blocking region with gate circuit open, **T**-on-state, **R**-reverse, **F**-forward
- Second subscript denotes the operating values: **W-** working value, **R-** repetitive value, **S-** surge or non-repetitive value, **T-** trigger
- Third subscript: **M-** maximum value
- **G-** gate ratings, **A** –stands for anode, **AV -** stands for average

VOLTAGE RATINGS

- Indicate the values of maximum voltages that a thyristor can withstand without a breakdown of the junction area with gate circuit open.

(i) $\boldsymbol{V_{DWM}}$ – Peak working forward blocking voltage:– maximum forward-blocking voltage that a thyristor can withstand during its working

– it is maximum value of the sine voltage wave.

VOLTAGE RATINGS

(ii) $\boldsymbol{V_{DRM}}$ – Peak repetitive forward blocking voltage:– peak transient voltage that a thyristor can withstand repeatedly in it forward-blocking mode.

- this rating is specified at a maximum allowable junction temperature

with gate circuit open.

- is encountered when a thyristor is turned-off.
- During turn-off, an abrupt change in reverse recovery current is accompanied by a spike in voltage

VOLTAGE RATINGS

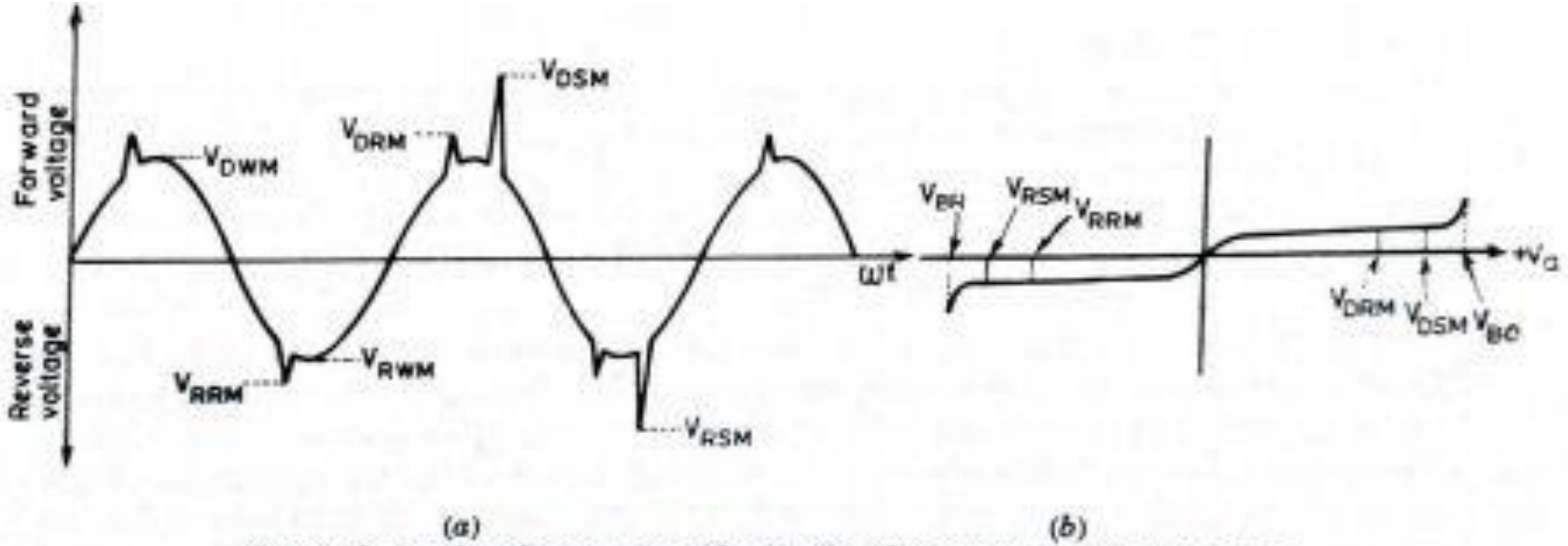


Fig. 4.17. Anode voltage ratings during the blocking state of a thyristor.

VOLTAGE RATINGS

(iii) $\boldsymbol{V_{DSM}}$ – Peak surge (or non-repetitive) forward blocking voltage: – peak value of the forward surge voltage that does not repeat

- about 130% of V_{DRM}
- $V_{DSM} < V_{BO}$.

(iv) $\boldsymbol{V_{RWM}}$ – Peak working reverse voltage:

– maximum reverse voltage that a thyristor can withstand repeatedly

– peak negative value of a sine voltage wave.

v) $\boldsymbol{V_{RRM}}$ – Peak repetitive reverse voltage:

– peak reverse transient voltage that may occur repeatedly in the reverse

- direction at the allowable maximum junction temperature

-lasts for a fraction of the time of one cycle.

VOLTAGE RATINGS

(vi) $\boldsymbol{V_{RSM}}$ – Peak surge (or non-repetitive) reverse voltage:

– peak value of the reverse surge voltage that does not repeat

- about 130% of V_{RRM}

– $\boldsymbol{V_{RSM}}$< V BR .

- $\boldsymbol{V_{DSM}}$ and $\boldsymbol{V_{RSM}}$ ratings can be increased by connecting a diode in series with a thyristor.
- Ratings (i) to (iii) – related to forward blocking voltages & ratings (iv) to (vi) – related to reverse blocking voltages.

(vii) $\boldsymbol{V_T}$ - On-state voltage drop:

– voltage drop between anode and cathode with specified on-state current and junction temperature

– of the order of 1 to 1.5 V

VOLTAGE RATINGS

(viii) Forward dv/dt rating of a thyristor:

- maximum rate of rise of the anode voltage that will not trigger the device without any gate signal
- if dv/dt is more than specified the value, thyristor may be switched on
- With forward voltage, J_2 is reverse biased
- Charges across J_2 develop a capacitance C.
- When a forward voltage is suddenly applied to the device, a charging current flows
- It may act as the gate current to turn-on the SCR without a positive gate signal.
- This method of triggering is not used as it gives random turn-on of a thyristor.
- This method also leads to damage of thyristor due to high junction temperature.

CURRENT RATINGS

- A thyristor – made up of semiconductor material – so thermal capacity is small
- junction temperature is dependent on the current handled by a thyristor
- even for short over currents the junction temperature may exceed rated value and the thyristor may get damaged
- Proper selection of the current rating is essential for a long working life of the device.

CURRENT RATINGS

(i) I_{TAV} -Average on-state current:

- it is the allowable average current that can be applied safely such that maximum junction temperature and rms current limit are not exceeded.
- the forward voltage drop across SCR during conduction is low and the power loss in a thyristor mainly depends on the average on-state current.
- for the junction temperature to remain within the limits, SCR must be rated at a value lower than I_{TAV} .
- repetitive in nature.
- this current determines the application of the device
- Dependent on maximum junction temperature
- With better cooling this rating can be improved

CURRENT RATINGS

(ii) I_{rms} – rms current:

- heating of resistive elements of a thyristor depends on forward rms current.

- this rating is used as the upper limit of anode current rating of the thyristor.

- Repetitive in nature.

- Dependent on maximum junction temperature

- With better cooling this rating can be improved

CURRENT RATINGS

(iii) Surge current rating:

- Maximum possible non repetitive or surge current which the device can withstand.

- When a thyristor is working below its repetitive voltage and current rating, its permissible junction temperature is never exceeded.

- Due to faults or short circuits, the thyristor may be subjected to abnormal operating conditions.

- Higher currents caused by non repetitive faults or short circuits should occur once in a while during the life span of the thyristor to prevent its degradation.

- Surge currents are sine waves with a frequency of 50 or 60 Hz, depending on the supply frequency

-This rating specified in terms of the number of surge cycles with corresponding surge current peak.

- this rating is inversely proportional to the duration of the surge

CURRENT RATINGS

(vi) Other ratings:

- Latching & holding currents
- Turn-on & turn-off times
- Gate circuit voltage, current and power ratings

CURRENT RATINGS

(v) di/dt rating:

- indicates the maximum rate of rise of current from anode to cathode

- when a thyristor is turned on conduction starts near the gate and this small area of conduction spreads to the whole area of the junction.

- If the rate of rise of anode current (di/dt) is large as compared to spread velocity of carriers across the cathode junction local hot spots will be formed near the gate due to high current density.

- This causes the junction temperature to rise above safe value and SCR will be damaged permanently.

- Therefore a limit on the value of di/dt at turn-on is specified.

- typical values of di/dt are 20 to 500 A/μsec.

Thyristor protection

- For reliable operation of SCR, it should be operated within the specific ratings.
- SCRs are very delicate devices and so they must be protected against abnormal operating conditions.
- Various protection of SCR are
    - di/dt protection
    - dv/dt protection
    - Over voltage protection
    - Over current protection
    - Gate protection

di/dt protection

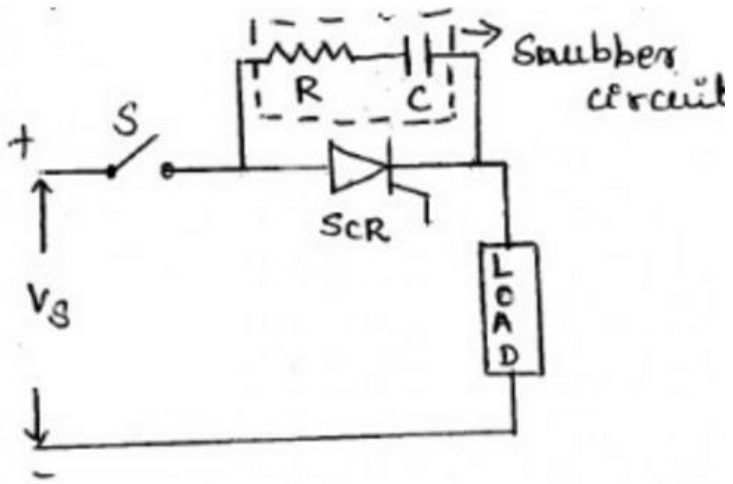
- If di/dt is large as compared to the spread velocity of carriers, local hot spots will be formed near the gate on account of high current density
- localised heating may destroy SCR
- The value of di/dt can be maintained below specified limit by using small inductor called di/dt inductor
- Typical di/dt limit value of SCR – 20-500 A/µs
- Gate current can be applied near to specified maximum value to avoid local spot heating

dv/dt protection

- During forward blocking mode, J2 has the characteristics of a capacitor
- Charging current, i_C=C.dv/dt. ie, when dv/dt is high, i_C is high.
- This high current may turn ON SCR even when gate current is zero. This is called as dv/dt turn ON or false turn ON of SCR.
- To protect the thyristor against false turn ON or against high dv/dt a "Snubber Circuit" is used.

Snubber circuit

- The snubber Circuit is a series combination of resistor 'R' and capacitor 'C'.
- They are connected across the thyristor to be protected.
- The capacitor 'C' is used to limit the dv/dt across the SCR.
- The resistor 'R' is used to limit high discharging current through the SCR.
- When switch S is closed, the capacitor 'C' behaves as a short-circuit.
- Therefore voltage across SCR is zero.
- As time increases, voltage across 'C' increases at a slow rate.
- Therefore dv/dt across 'C' and SCR is less than maximum dv/dt rating of the device.
- The capacitor charges to full voltage Vs; after which the gate is triggered, and SCR is turned ON and current flows through SCR.



SCR – TURN-ON METHODS

TURN-ON METHODS

SCR can be turned on by:- (provided the anode should be +ve wrt cathode)

1. Forward voltage triggering
2. Gate triggering
3. Temperature triggering
4. Light triggering
5. *dv/dt* triggering

TURN-ON METHODS (Contd….)

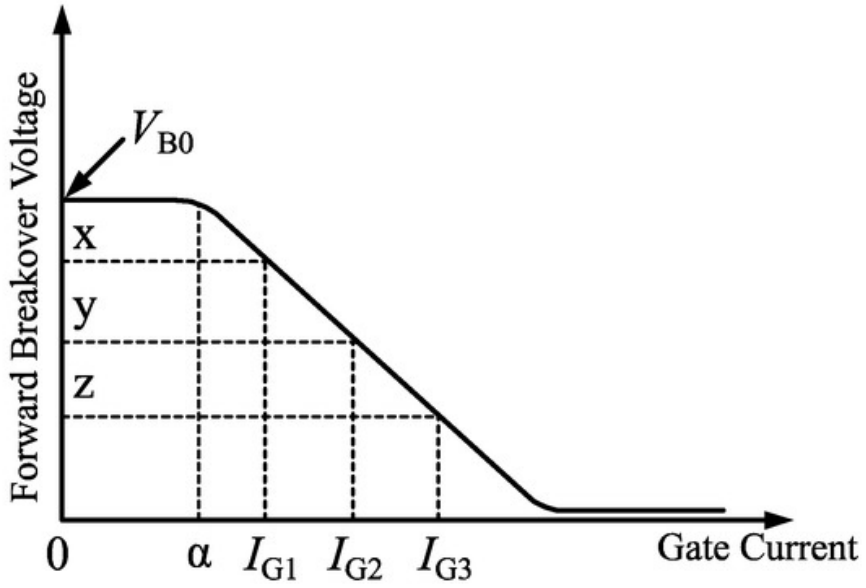
Gate triggering:-

- Common method for firing forward biased SCRs
- Most simple, reliable & efficient method
- A thyristor will remain in forward blocking mode with normal working voltage and gate circuit open
- To turn-on the thyristor, a +ve gate voltage is applied between Gate & Cathode
- Gate starts to flow, charges are injected into the inner p-region, this reduces the voltage at which the forward breakover occurs

TURN-ON METHODS (Contd….)

Gate triggering:- (Contd...)

- Gate current determines the forward breakover voltage at which the device switches from off-state to on-state.
- Higher the gate current, lower will be the breakover voltage.
- When positive gate signal is applied, inner p-layer is flooded with electrons from cathode
- This is because outer n-layer is heavily doped than inner p-layer.



GATE CHARACTERISTICS

- Gate-cathode circuit is similar to a *p*-*n* diode
- The V-I chara of gate is also similar to that of a diode
- The circuit which supplies firing signals to the gate must be designed

  - to assure triggering at the desired instant

  - not to exceed the maximum voltage & power capabilities of the gate

  - to prevent triggering from false signals or noise

GATE CHARACTERISTICS

- Figure below shows the trigger circuit of an SCR

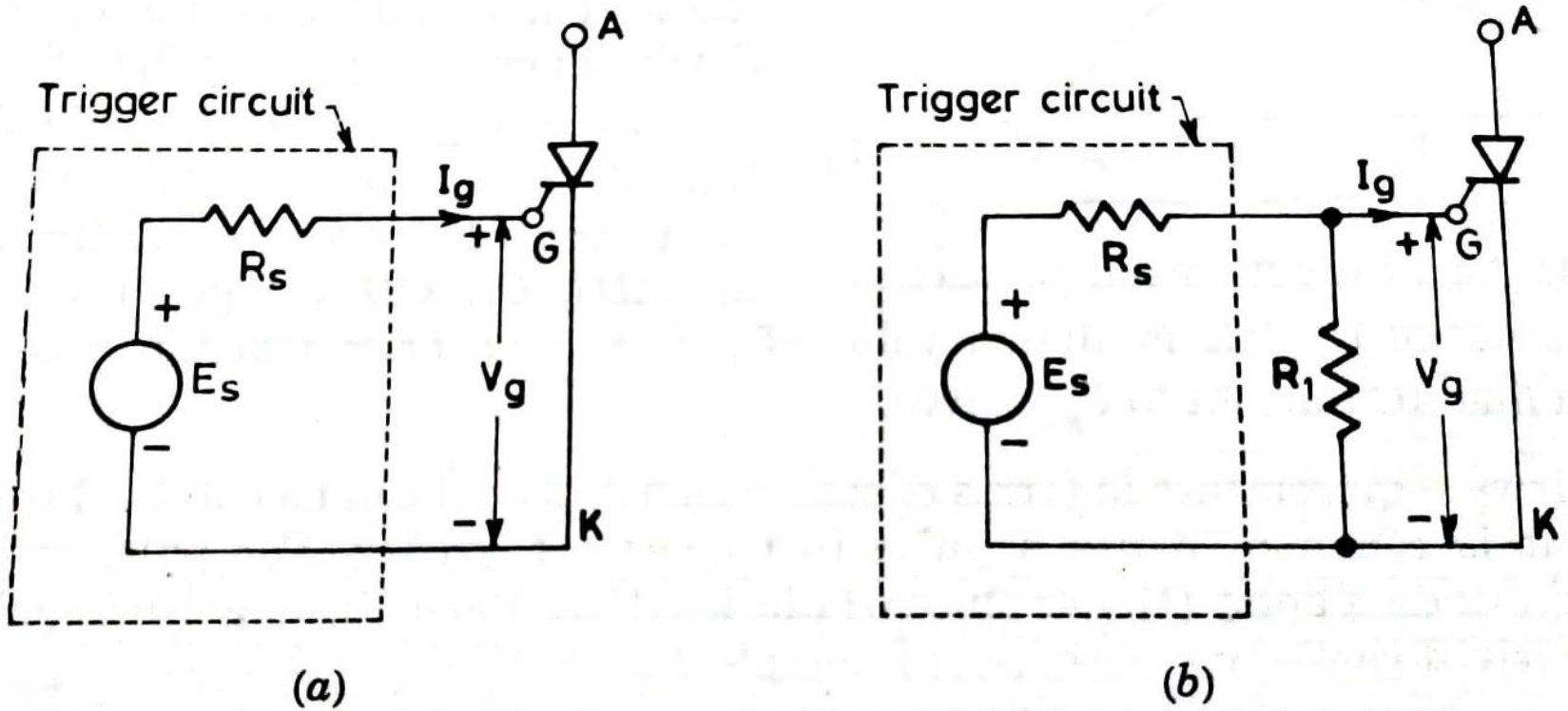
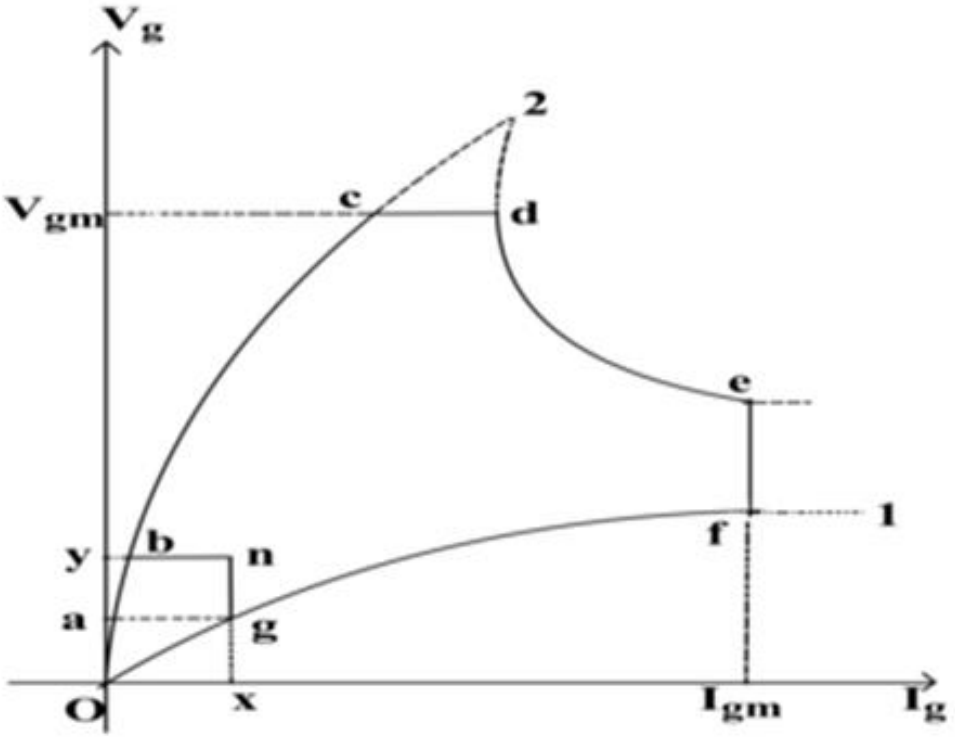
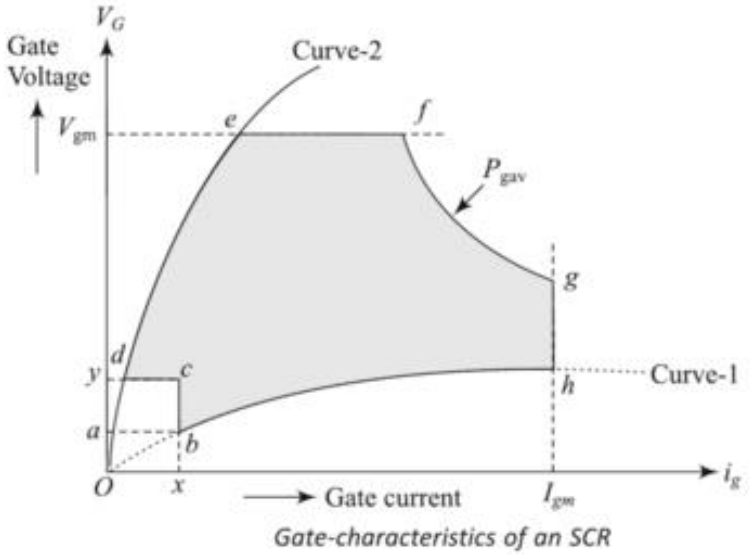


Fig. Trigger circuit connected to gate-cathode circuit of an SCR.

GATE CHARACTERISTICS (V_g-I_g chara)

- Forward gate chara- plot between gate voltage(V_g) and gate current(I_g)
- V_g and I_g represent DC values
- Applying gate drive increases the minority carrier density in the inner *p-layer* , this facilitate the reverse breakdown of junction J_2.





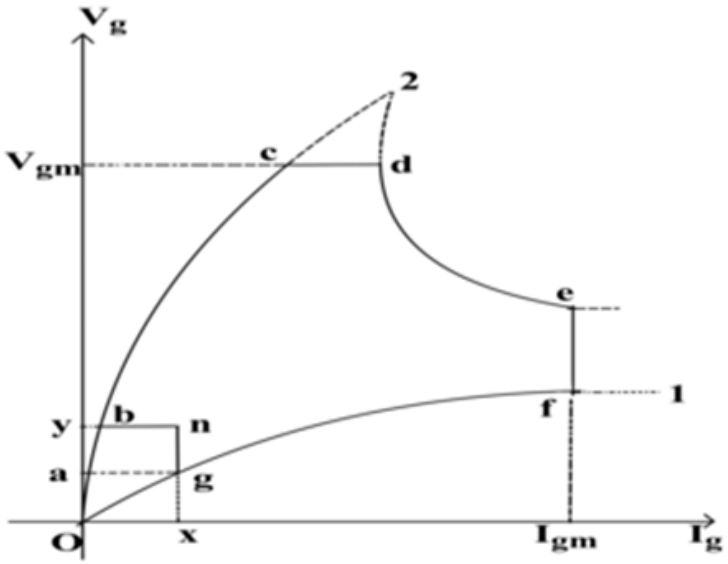
Gate-characteristics of an SCR

GATE CHARACTERISTICS (V_g-I_g chara)

- V_g - I_g chara has a spread between curves 1 & 2
- Spread is due to the doping density P_2 and N_2 layers.
- Curve 1 represents the lowest voltages that must be applied to turn-on the SCR
- Curve 2 represents the highest voltages that can be safely applied to the gate circuit of the SCR.
- Each thyristor has maximum limits as V_{gm}(for gate voltage) and I_{gm} (for gate current) and rated average gate power dissipation P_{gav}
- These maximum values prevent the permanent damage of junction J_3 and provide reliable triggering.
- There are minimum levels for V_g and I_g for reliable turn-on.
- Thyristor can be destroyed if V_{gm}, I_{gm} *and* P_{gav} are exceeded

GATE CHARACTERISTICS (V_g-I_g chara)

- A non-triggering voltage is also specified by manufacturers.
- In the fig oa is the non-triggering voltage
- If the triggering circuit generates an unwanted signal, it's magnitude should be less than the non-triggering voltage to prevent unwanted triggering of SCR.
- All noise signals should be less than non-triggering voltage.



GATE CHARACTERISTICS (V_g-I_g chara)

- From fig (a) $E_s = V_g + I_g R_s$

where E_s = gate source voltage

V_g = gate-cathode voltage and

R_s = gate-source resistance

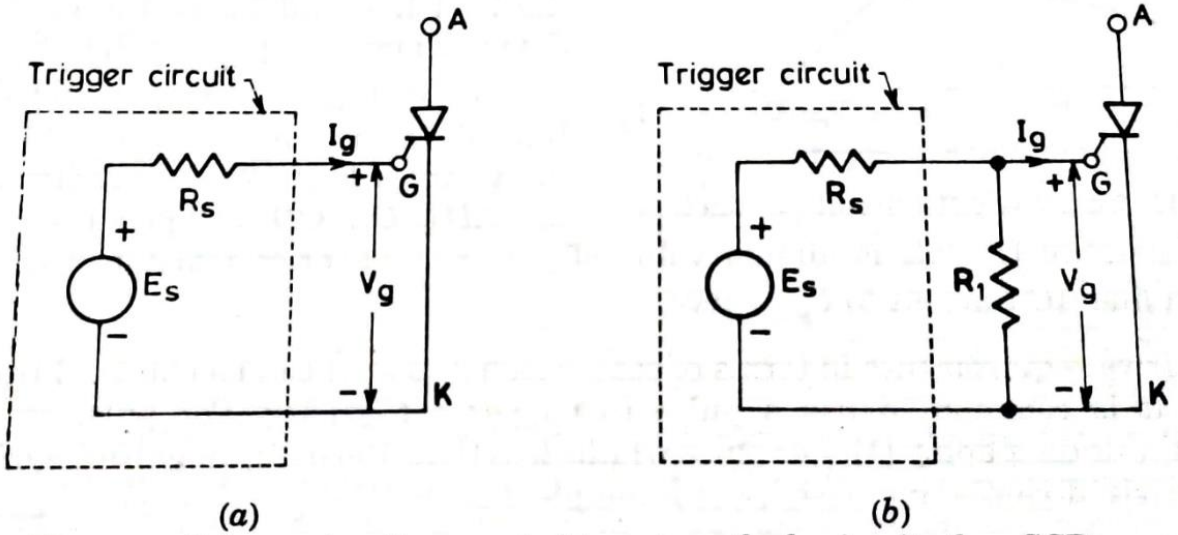


Fig. Trigger circuit connected to gate-cathode circuit of an SCR.

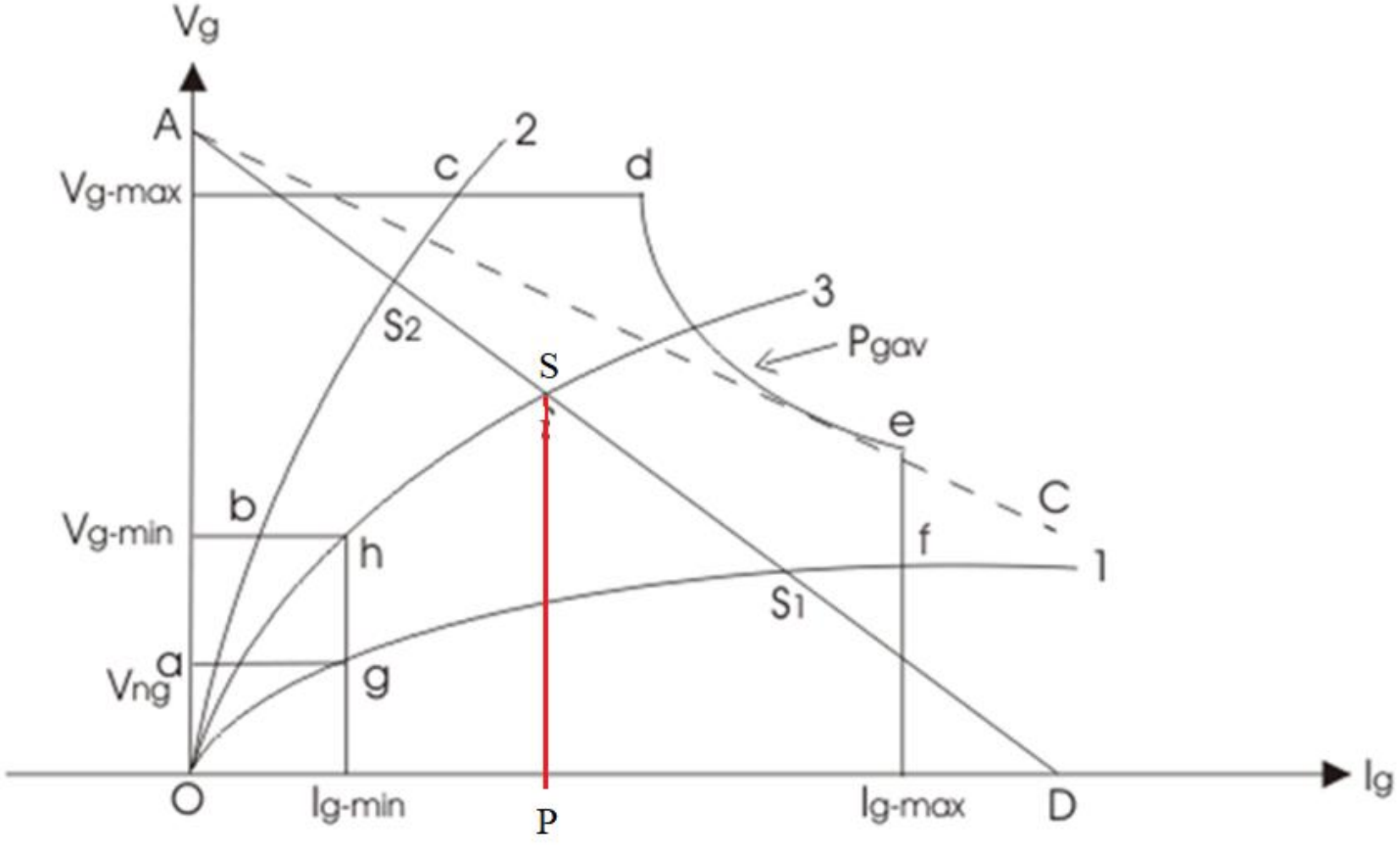
GATE CHARACTERISTICS (V_g-I_g chara)

- R_s is selected such that the current (= E_s/R_s) through it should not cause damage to the source and the gate circuit.
- If R_s is low, an external resistance should be added in series with it
- To provide an easy path to the flow of leakage current between SCR terminals a resistance R_1 is connected across the gate-cathode terminals.
- Let V_{gmn} and I_{gmn} be the minimum gate voltage and gate current to turn-on the SCR, then

current through $R_1 = V_{gmn}/ R_1$

and source voltage $E_s = V_{gmn} + (I_{gmn}+V_{gmn}/ R_1) R_s$

GATE CHARACTERISTICS (V_g-I_g chara)



GATE CHARACTERISTICS (V_g-I_g chara)

- In the fig OA = E_s , the gate source voltage and OD = trigger short circuit current = E_s / R_s
- Consider a thyristor with a chara given by curve 3
- Operating point is given by the intersection of curve 3 and the load line.
- For this SCR, gate voltage = PS & gate current = OP
- To minimise the turn-on time and unreliable turn-on the load-line and the operating point must be as close to the P_{gav} curve as possible

GATE CHARACTERISTICS (V_g-I_g chara)

- The operating curves must lie within the limit curves 1 & 2
- Gradient of load line = required gate source resistance
- Tangent to P_{gav} curve – minimum value of gate source resistance
- Gate-drive requirements in terms of continuous signal DC signal can be obtained from this V-I chara.
- Commonly a pulse is used to trigger a thyristor

GATE CHARACTERISTICS (V_g-I_g chara)

- Thyristor is a charge controlled device
- Higher the magnitude of gate current pulse, lesser is the time to inject the required to inject charge for turning-on the thyristor
- Turn-on time can be reduced by using gate current of higher magnitude.
- The gate drive signal can be DC, AC or a sequence of high frequency pulses.
- Gate pulse width should be sufficient to allow the anode current to exceed latching current
- Gate pulse width normally taken as ≥ turn-on time of SCR
- i.e., $T \geq t_{on}$

GATE CHARACTERISTICS (V_g-I_g chara)

- With pulse triggering, greater amount of power dissipation can be allowed.
- But this power dissipation should be less than the peak instantaneous gate power dissipation $P_{gm.}$
- Frequency of firing pulses in Hz, $f = 1/T_1 = P_{gav}/(T.P_{gm})$

where T – pulse width in sec, T_1 – periodicity of the pulse

- Duty-cycle (δ) – ratio of on-period of the pulse to the periodic time of the pulse
- $\delta = T / T_1 = fT$
- $P_{gav} / \delta = P_{gm}$

GATE CHARACTERISTICS (V_g-I_g chara)

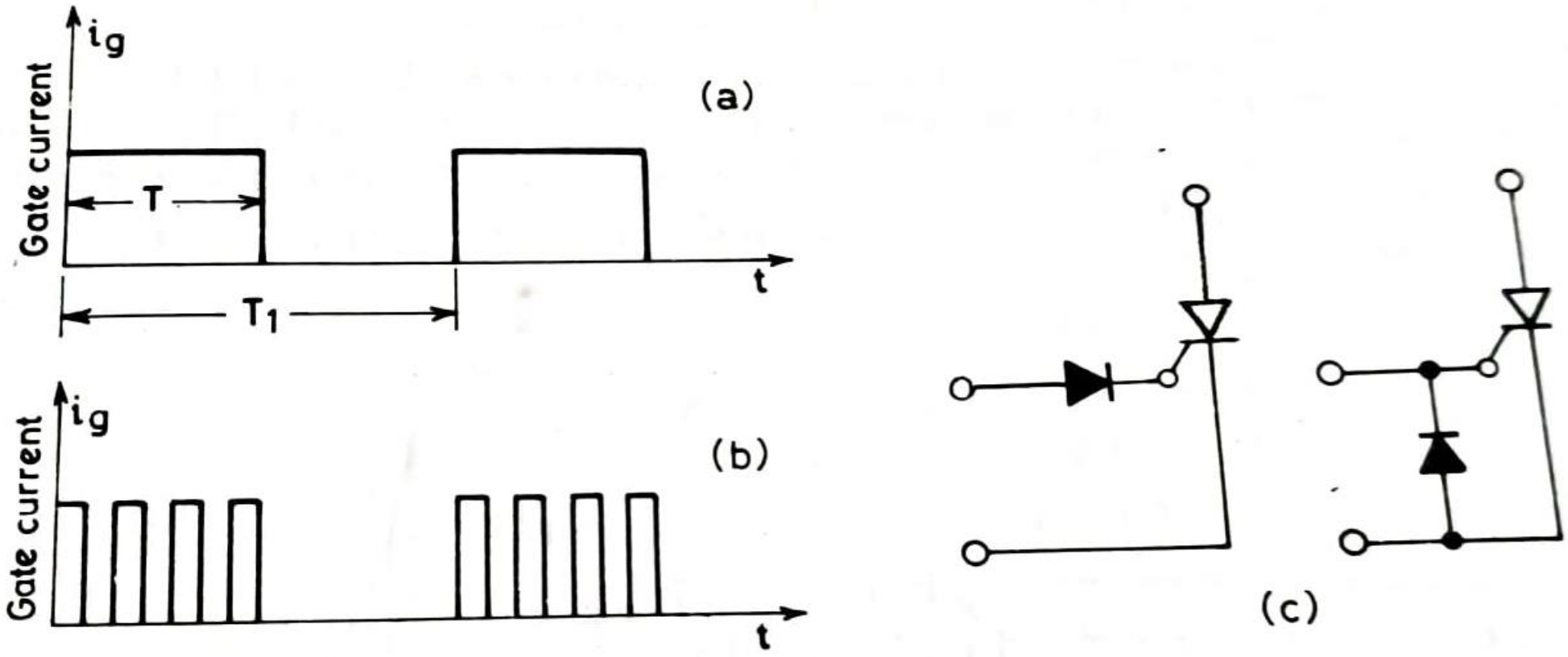


Fig. (*a*) Pulse gating and (*b*) high-frequency carrier gating of SCRs, (*c*) Thyristor protection against reverse overvoltages.

GATE CHARACTERISTICS (V_g-I_g chara)

- Sometimes the pulses are modulated to generate a train of pulses
- This technique is called **high frequency carrier gating**
- **Advantages** – lower rating, reduced dimensions, overall economical design of the pulse transformer needed for isolating the low power circuit from the main power circuit.
- For an SCR V_{gm} and I_{gm} are specified separately
- Pulse voltage amplitude x pulse current amplitude < P_{gm}

Gate Triggering Circuits

Gate Triggering Circuits

1. Resistance (R) Triggering Circuit
2. Resistance-Capacitance (RC) Triggering Circuit
3. UJT Based Triggering Circuit

R Triggering Circuit

- Simplest triggering circuit
- Triggering angle range (0 to 90^0)

R_1- Current limiting resistor

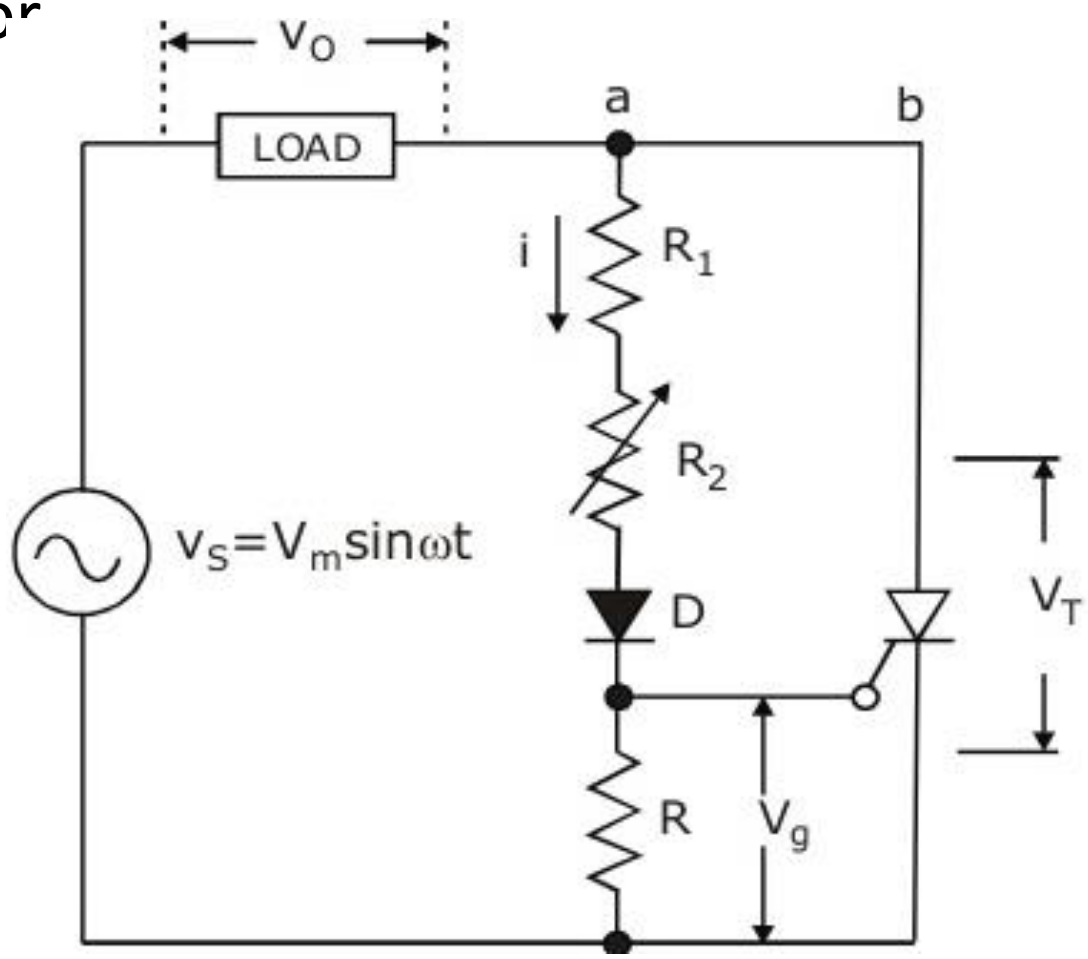
R_2 – Variable resistor

R – Stabilizing resistor

I_{gm} – max. permissible gate current

V_{gm} - max. permissible gate voltage

V_m – max. value of source voltage



$$R_1 \geq \frac{V_m}{I_{gm}}$$

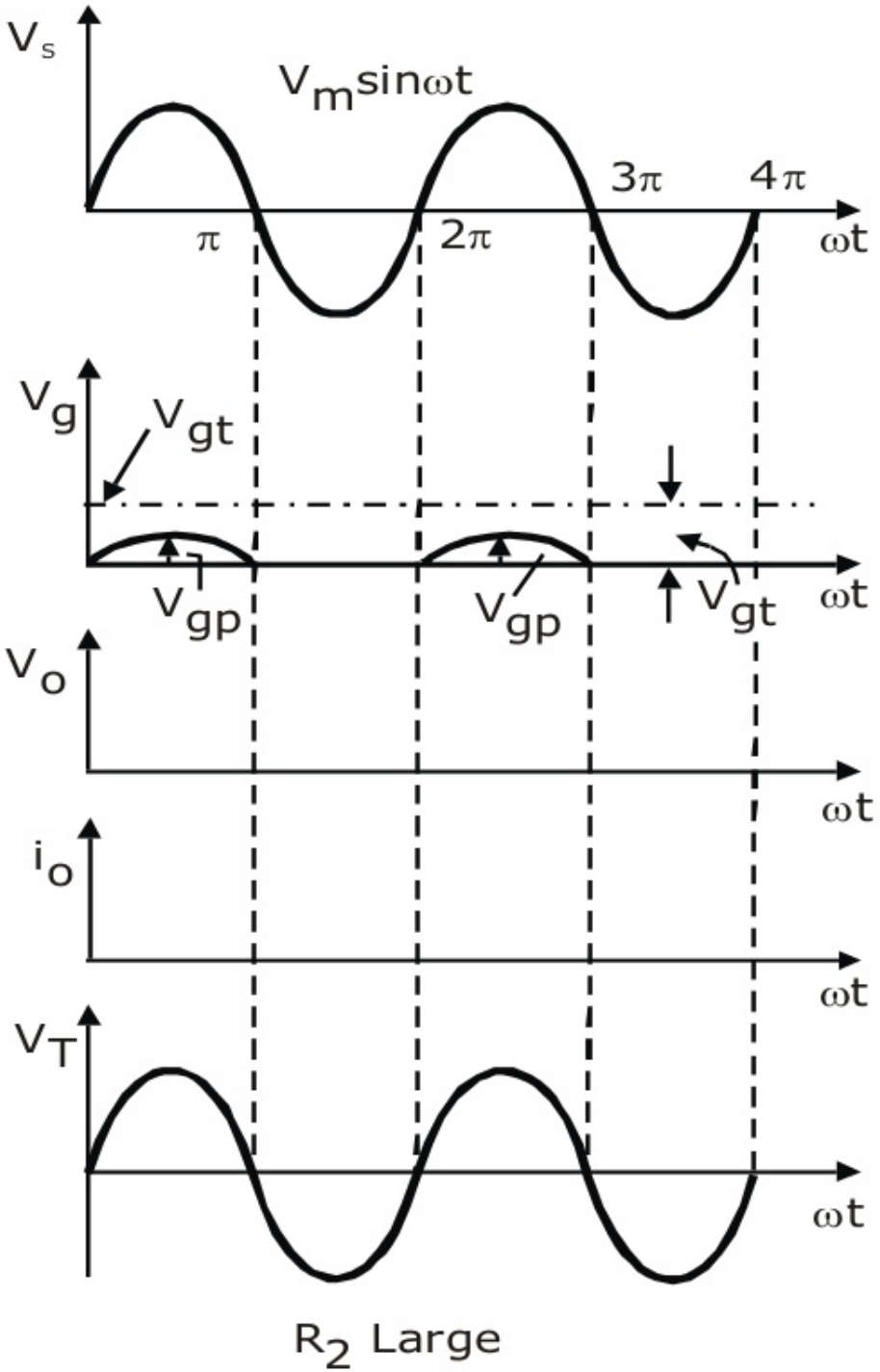
$$\frac{V_m}{R + R_1} R \leq V_{gm} \Rightarrow R \leq \frac{V_{gm} R_1}{V_m - V_{gm}}, \quad (R_2 = 0)$$

- When $V_{gp} > V_{gt}$, $\alpha < 90°$
- Also α can't be zero however large v_g may be.
- Min. value of α is about 2° – 4° (which is obtained when $R_2 = 0$)
- **Relationship between V_{gp} & V_{gt} is**

$$V_{gp} \sin\alpha = V_{gt}$$

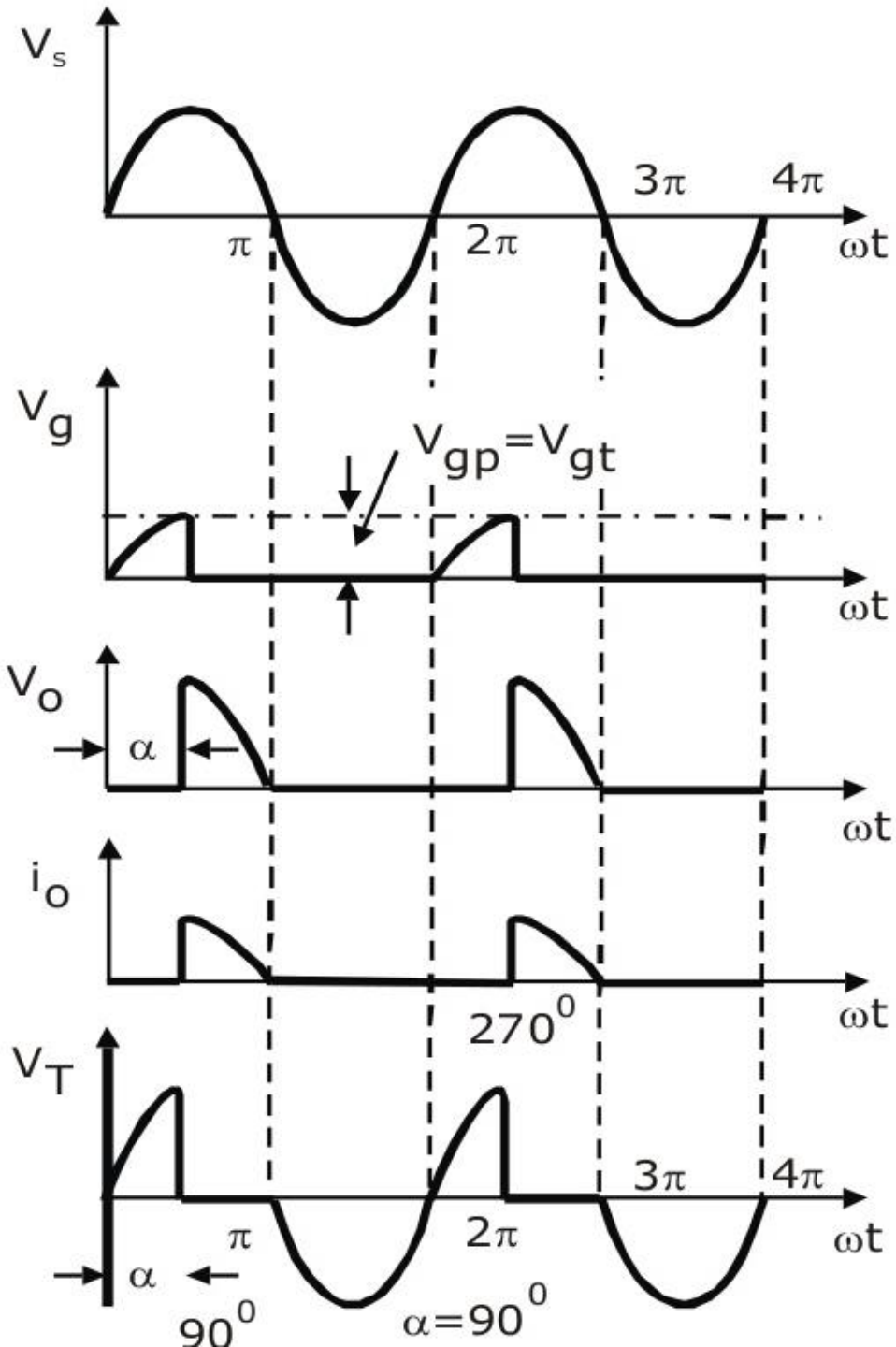
$$\text{or, } \alpha = \sin^{-1} (V_{gt} / V_{gp})$$

No Triggering

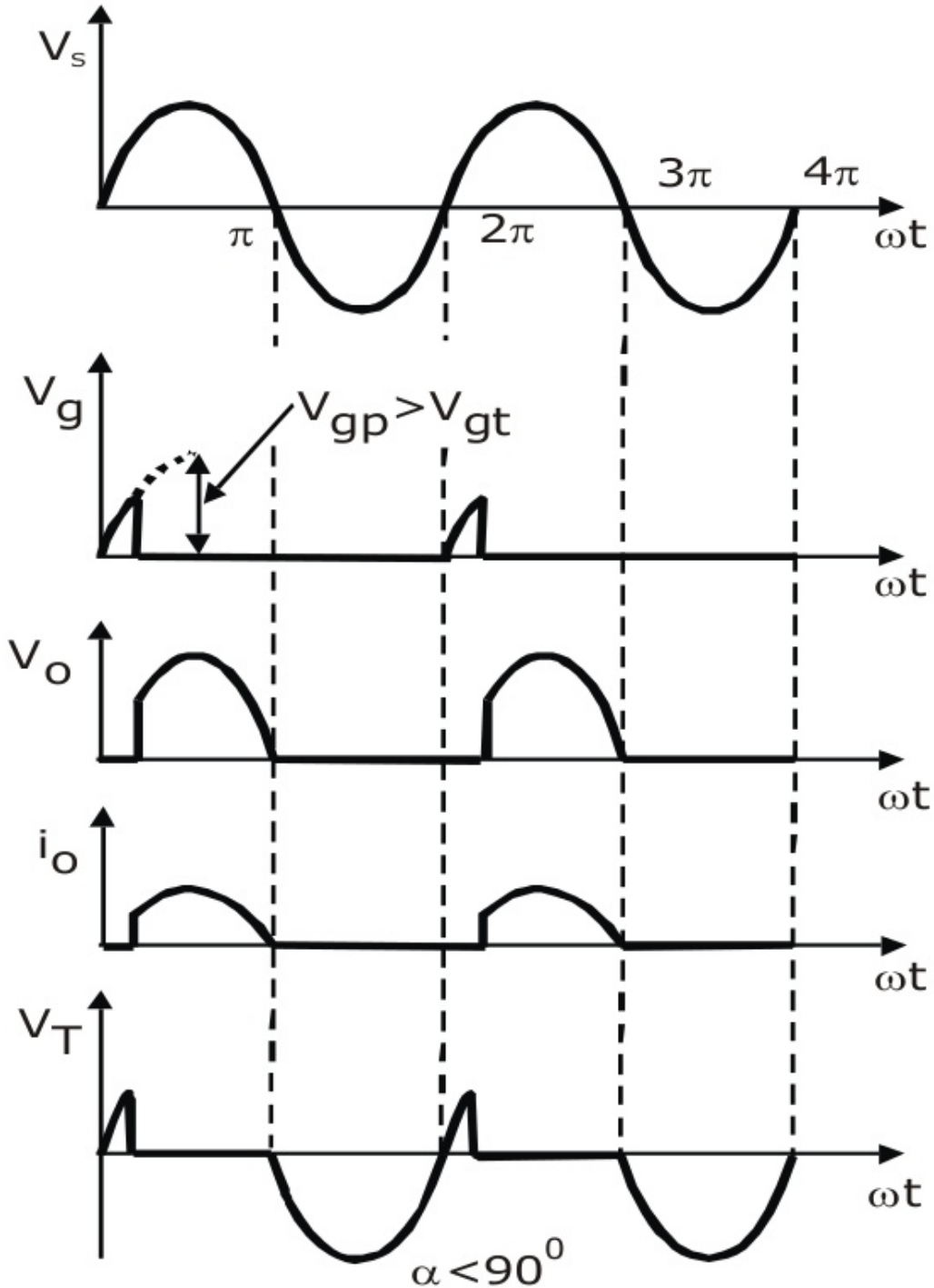


R_2 Large

α = 90°

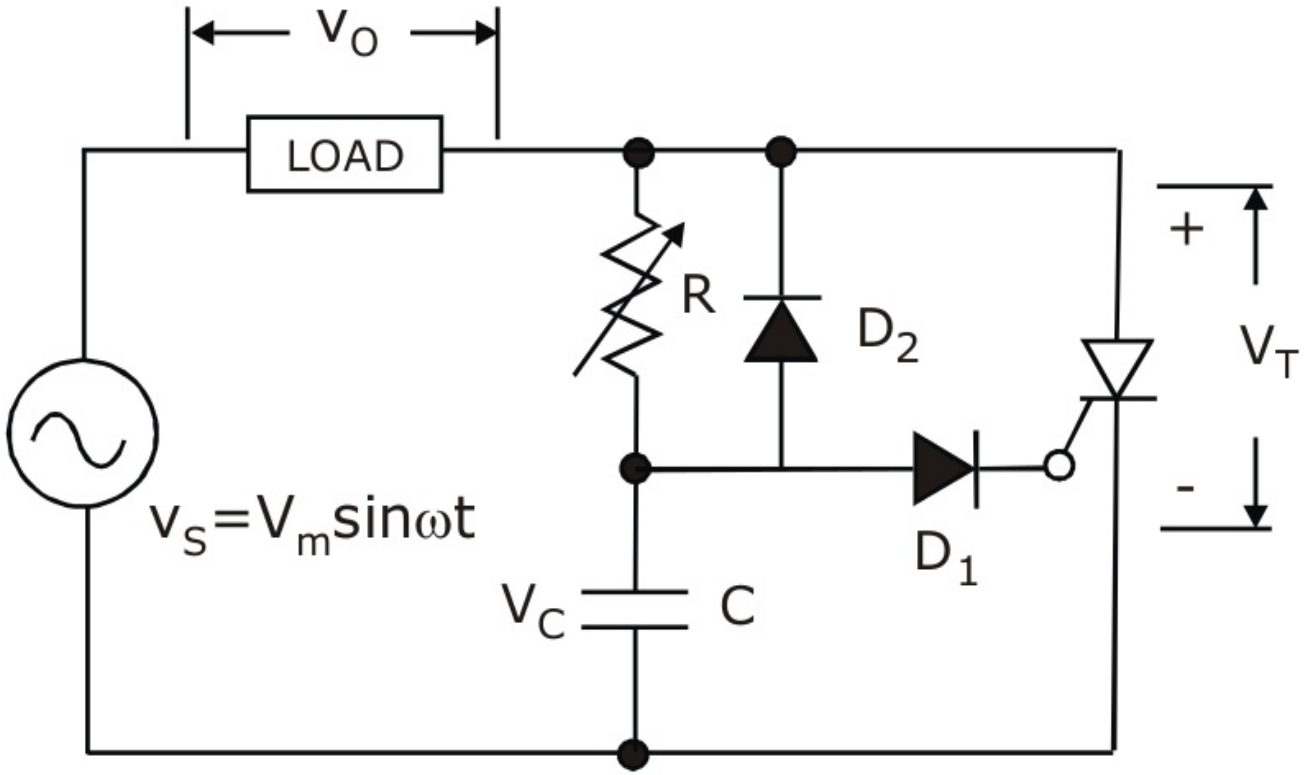

V_s
3π
4π
π
2π
ωt
V_g
$V_{gp}=V_{gt}$
ωt
V_o
α
ωt
i_o
270^0
ωt
V_T
3π
4π
π
2π
ωt
α
90^0
$\alpha=90^0$

α < 90°



RC triggering circuit

Triggering angle range (0 to 180^0)



$$RC \geq \frac{1.3T}{2}$$

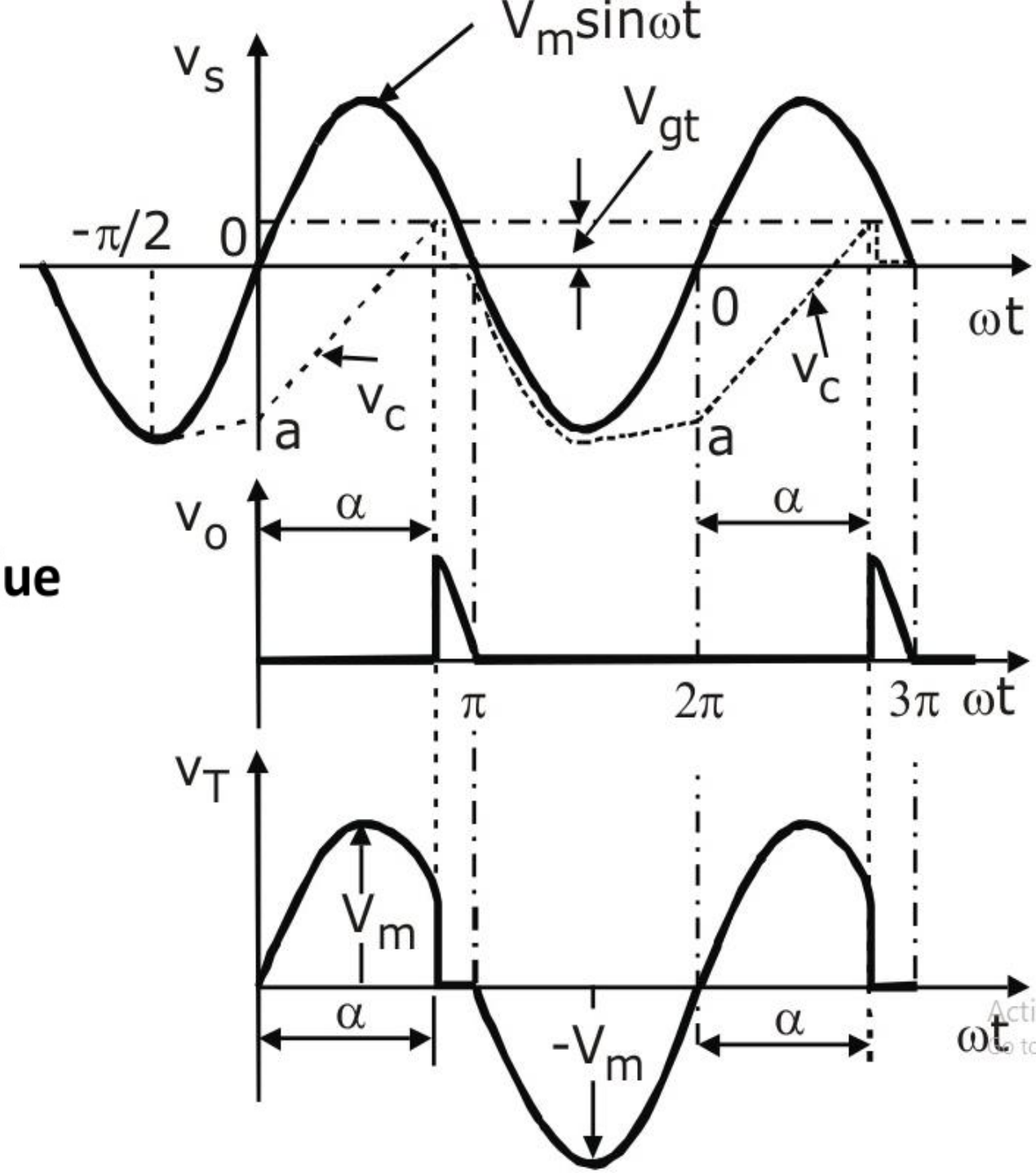
T=1/f

At the triggering instant, $v_C = V_{GT} + V_{D1}$

Max value of R can be found as follows

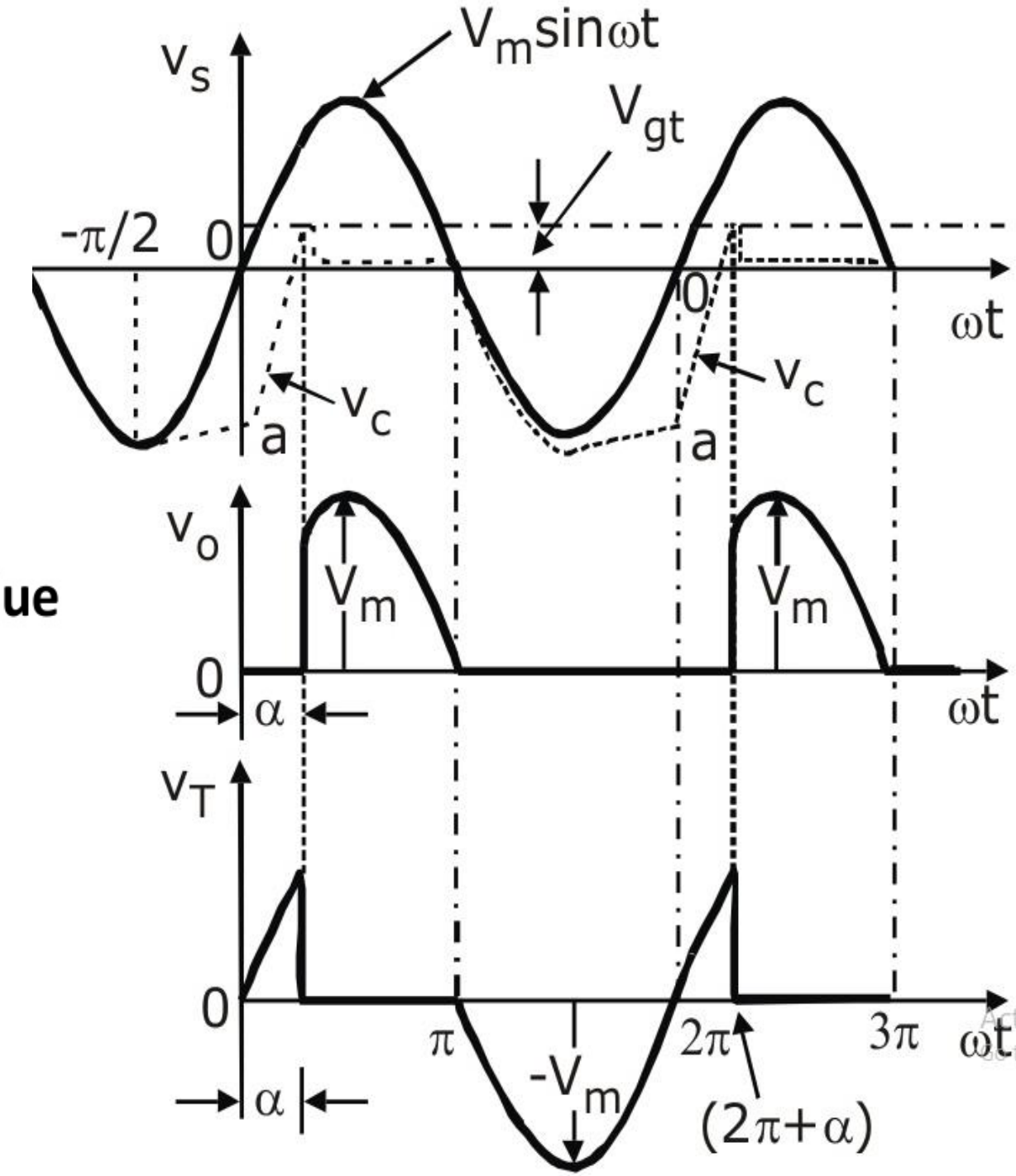
$$R \leq \frac{v_s - V_{GT} - V_{D1}}{I_{GT}}$$

High Value of R



Activat
Go to Se

Low Value of R



Activate
Go to Setti

Gate Pulse Magnitude (Current and Voltage)

Minimum Requirement:

- The gate current must be greater than the minimum gate current required to trigger the SCR for reliable turn-on.
- Similarly, the gate voltage must exceed the minimum gate trigger voltage.

Optimal Magnitude:

- The amplitude should typically be 3 to 5 times the minimum I_{GT} to achieve rapid and uniform turn-on across the entire cathode junction, a process called "hard firing" or "overdriving".

Maximum Limit:

- The gate current and voltage must not exceed the maximum allowable gate power dissipation P_{GM} or surge current limits specified (Data sheet).

Gate Pulse Duration (Width)

- Latching Condition: The primary requirement for pulse duration is that it must be long enough for the anode current to rise and reach the latching current I_L of the device before the gate pulse is removed.
- If the pulse is removed too soon, the SCR will turn off.

Typical Duration:

- For a simple resistive load, the latching may happen quickly.
- For inductive loads, the anode current rises more slowly, requiring a longer pulse.

- A typical duration for a single pulse can range from 50 µs to 200 µs (microseconds), and for some applications, even longer.

Gate Pulse Duration (Width)

Pulse Train for Inductive Loads:

- For highly inductive loads where the anode current takes longer to build up, a continuous DC signal or a "train of pulses" (a series of short, high-frequency pulses) may be used for the entire conduction period to ensure latching and continuous conduction.

Minimizing Gate Power Loss:

- Using a pulse or pulse train instead of a continuous DC signal after the device is latched helps reduce unnecessary gate power dissipation and heat generation.

Gate Pulse Duration (Width)

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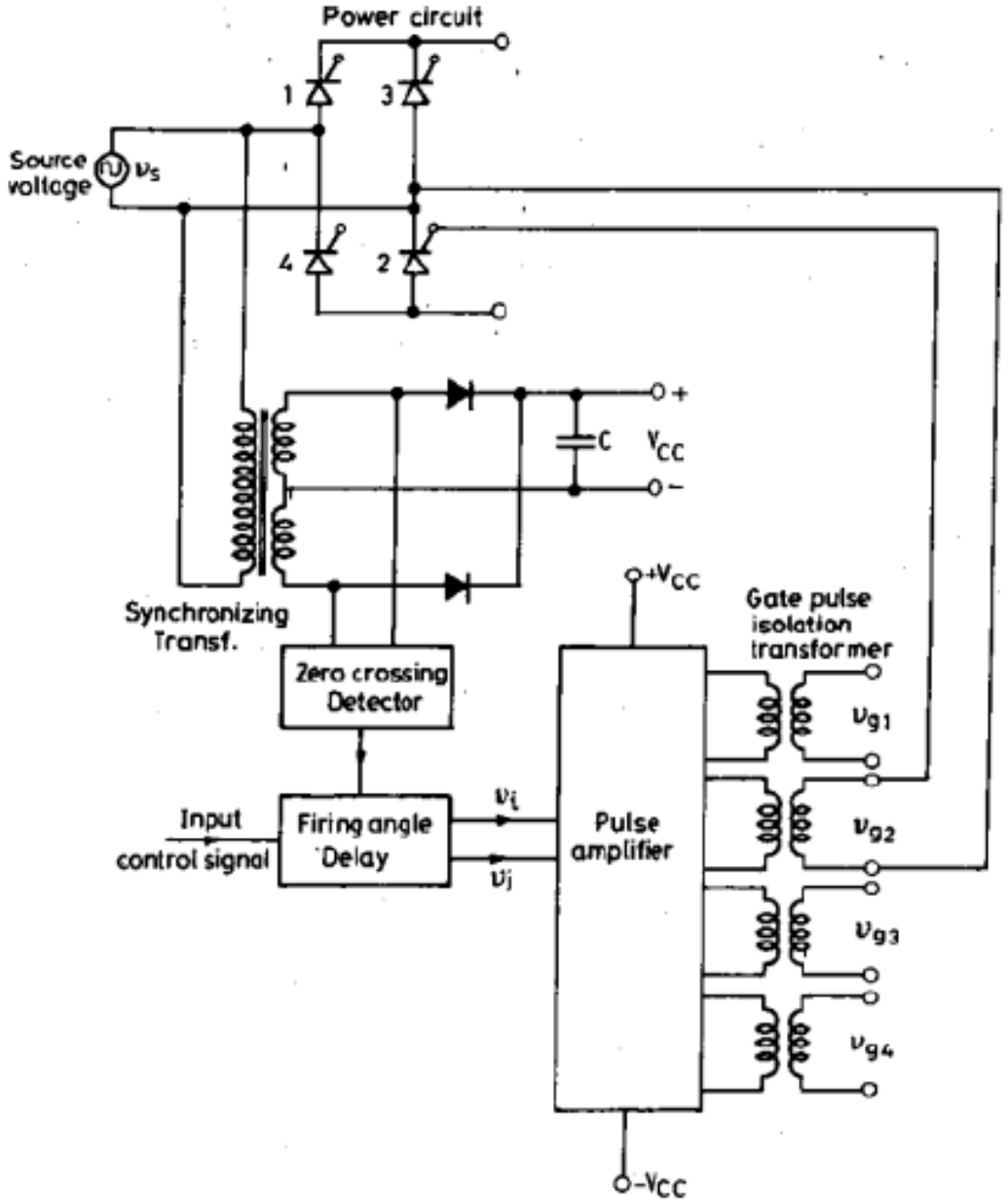
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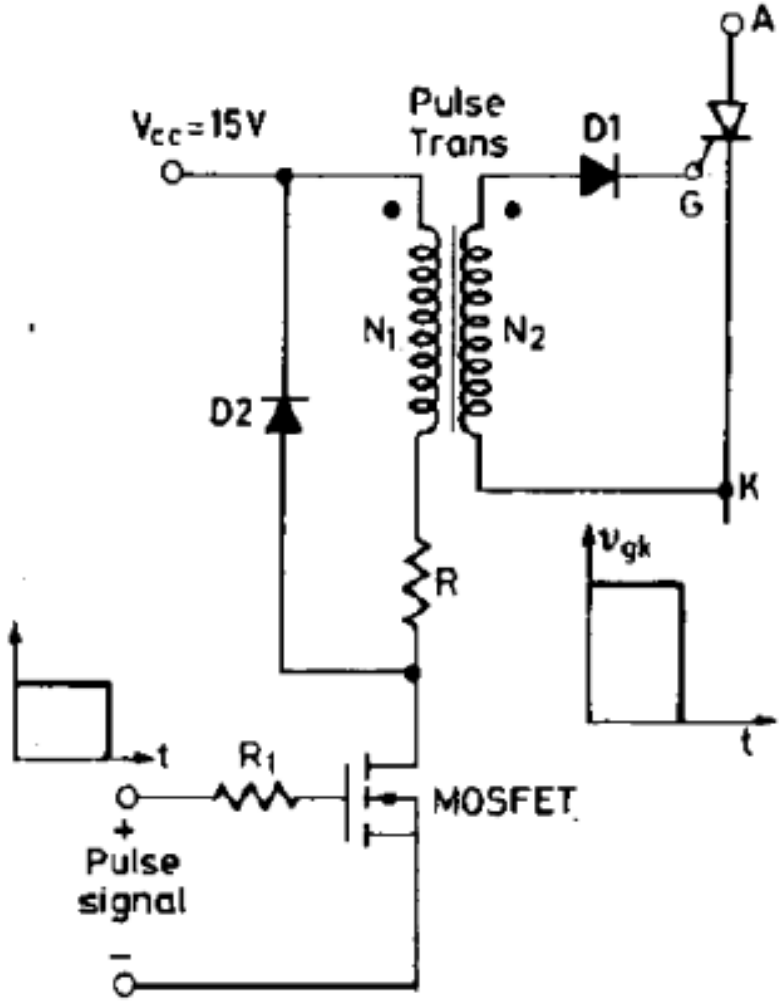
Gate synchronization of a Silicon Controlled Rectifier (SCR)

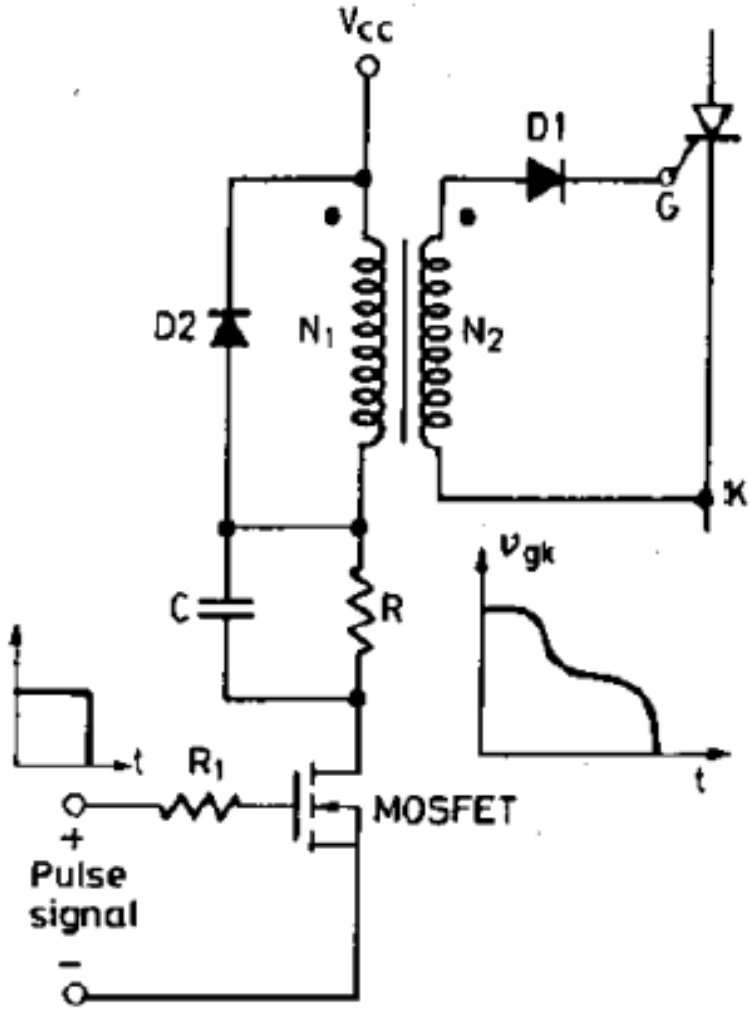
- The process of precisely timing the application of its gate trigger pulse with the alternating current (AC) power supply waveform is called gate synchronization.
- Allows the user to determine the exact moment within each half-cycle that the SCR begins to conduct.
- This synchronization is essential for controlled power regulation in AC circuits, such as in rectifiers and motor controls.

Typical SCR trigger circuit



Short pulse and Long pulse gating





Pulse train gating

